

Uncertainty quantification of planetary entry technologies

Andrew J. Brune^a, Thomas K. West IV^b, Serhat Hosder^{c,*}

^a Structural and Thermal Systems Branch, NASA Langley Research Center, USA

^b Vehicle Analysis Branch, NASA Langley Research Center, USA

^c Department of Aerospace and Mechanical Engineering, Missouri University of Science and Technology, USA

ARTICLE INFO

Keywords:

Planetary entry
Uncertainty quantification
Sensitivity analysis
Hypersonic flow
Thermal protection system
Fluid-structure interaction

ABSTRACT

The analysis, design, and development of planetary entry technologies rely heavily on computational modeling due to limited ground and flight test data, which simulate the extreme environments encountered by spacecraft during atmospheric entry. In general, these computational models possess uncertainties that may affect the prediction accuracy of the simulations. This makes uncertainty quantification a necessary tool for assessing the accuracy of model predictions and improving the robustness and reliability of entry systems. The objectives of this paper are twofold: to review uncertainty quantification studies applied to the modeling and simulation of planetary entry systems and to demonstrate an efficient uncertainty quantification approach based on stochastic expansions on the multidisciplinary analysis of a Hypersonic Inflatable Aerodynamic Decelerator configuration for Mars entry. The review of uncertainty quantification studies focuses on flight mechanics, guidance and navigation simulations, aerothermal modeling and prediction, fluid-structure interaction simulations, and thermal protection system response modeling of planetary entry technologies.

1. Introduction

The analysis, design, and development of planetary entry technologies rely heavily on computational modeling due to limited test envelopes of ground facilities in simulating extreme environments encountered by spacecraft during atmospheric entry. In general, these computational models possess uncertainties that affect the prediction accuracy when used in the design of the state-of-the-art planetary entry capsules with rigid aeroshells (e.g., the Mars Science Laboratory (MSL) spacecraft) as well as the new deployable technologies such the Hypersonic Inflatable Aerodynamic Decelerator (HIAD) [1,2] and the Adaptable Entry Placement Technology (ADEPT) [3] concepts which are currently being investigated for future human missions to Mars (Fig. 1) [4–6]. The importance of computational modeling in planetary entry technology development makes uncertainty quantification (UQ) a necessary tool for assessing the accuracy of model predictions and improving the robustness and reliability of the entry systems. The objectives of this paper are twofold: to give a review of UQ studies applied to the modeling and simulation of planetary entry systems and to demonstrate the implementation of an efficient UQ approach based on stochastic expansions on the multi-disciplinary analysis of a HIAD for Mars entry.

Uncertainties in the planetary entry simulations can arise due to lack of knowledge in the physical modeling and/or inherent variations in the physical system. As an example, Fig. 2 shows the complex aerothermodynamic environment around a generic planetary entry capsule which requires the simulation of high-temperature chemically-reacting laminar or turbulent shock-layer flow with non-equilibrium thermochemistry coupled with the thermo-mechanical response of an ablating thermal protection system (TPS). The reliability of TPS design depends on the accuracy of the simulations. These simulations are subjected to uncertainties in the modeling of various physical phenomena associated with the extreme aerothermodynamic environment and the inherent variations associated with the atmospheric conditions, speed, orientation, and geometry of the entry vehicle. To resolve the complex physics of planetary entry, high-fidelity computational models are frequently used which makes the use of traditional UQ methods based on sampling, such as Monte Carlo, infeasible due to enormous computational cost required for the convergence of the desired statistical metrics. Hence, besides the review of previous UQ studies on planetary entry technologies in general, this paper will also focus on the description of efficient and accurate UQ approaches for planetary entry simulations and demonstrate the implementation of such an approach on the multi-disciplinary analysis of a HIAD configuration, which includes high-

* Corresponding author.

E-mail addresses: andrew.j.brune@nasa.gov (A.J. Brune), thomas.k.west@nasa.gov (T.K. West), hosders@mst.edu (S. Hosder).

<https://doi.org/10.1016/j.paerosci.2019.100574>

Received 8 March 2019; Received in revised form 19 August 2019; Accepted 8 September 2019

Available online 11 October 2019

0376-0421/© 2019 United States Government as represented by the Administrator of the National Aeronautics and Space Administration. Published by Elsevier Ltd.

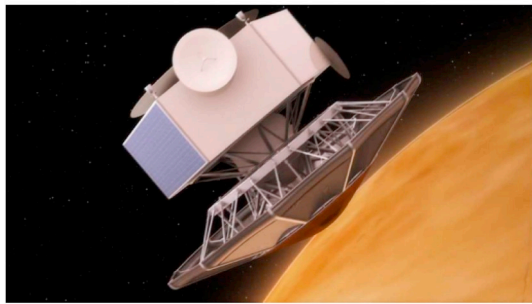
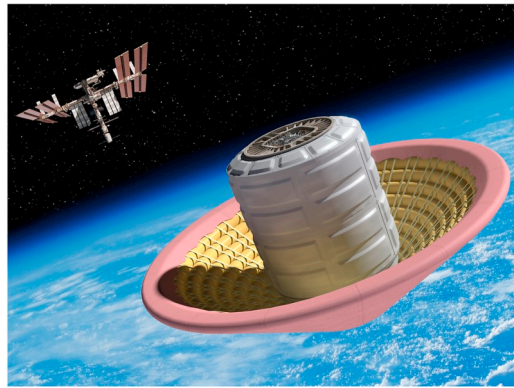
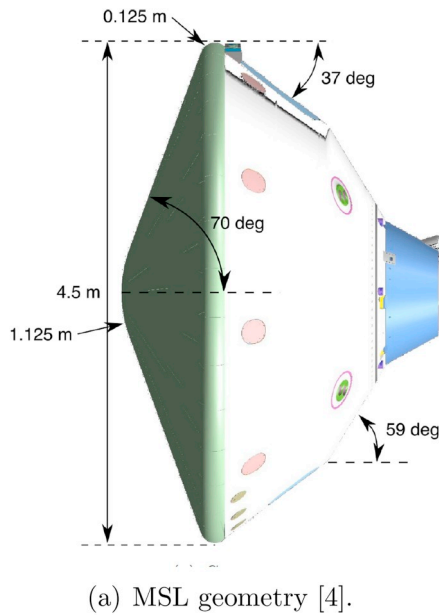


Fig. 1. Planetary entry technologies with rigid and deployable aeroshells.

fidelity computational aerothermodynamics of shock-layer flow, fluid-structure interaction, and flexible TPS modeling [7].

The integration of UQ into simulations can provide multiple benefits to the analysis, design, and development of planetary entry technologies (Fig. 3). In particular, UQ is an essential component of the validation of the computational models used in the simulations which requires the comparison of the numerical results to the ground or flight test data, all of which should be accompanied with well-quantified uncertainties for proper assessment and improvement of the physical models through validation metrics [8]. In addition, the sensitivity information obtained as a byproduct of the UQ process can be used to determine major sources contributing to the output uncertainty which can be used for resource

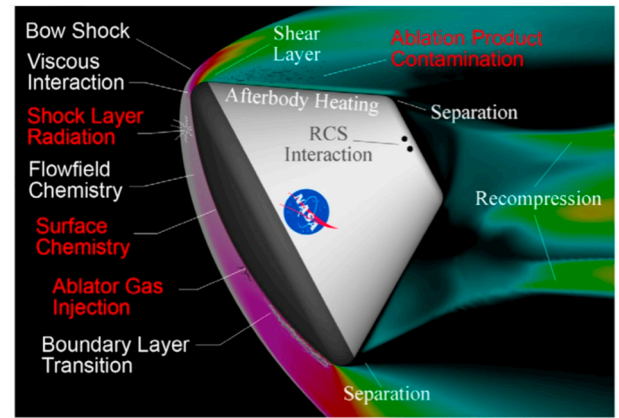


Fig. 2. Schematic outlining planetary entry aerothermodynamics over a rigid aeroshell capsule.

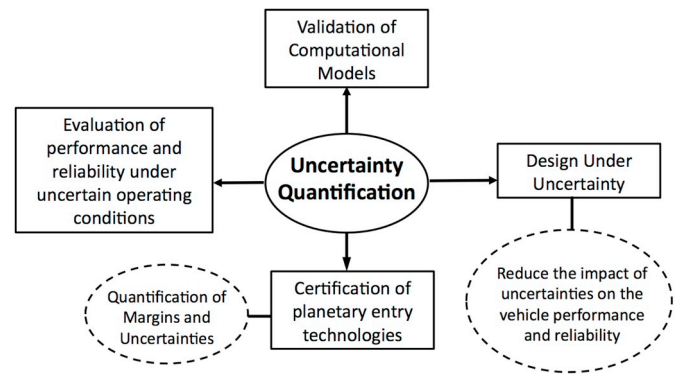


Fig. 3. Potential benefits of UQ for planetary entry technologies.

allocation for the planning and design of validation experiments to reduce uncertainty in critical physical models and parameters identified by sensitivity analysis [9]. Besides validation, UQ also enables the evaluation of performance and reliability of technologies under uncertain operating conditions such as the variations in the atmospheric conditions and speed of the vehicle during planetary entry. The direct integration of UQ into the design and optimization process can help reduce the impact of uncertainties on the performance of the entry system by increasing the robustness and reliability while optimizing the configurations to eliminate unnecessary design margins such as oversizing the TPS. Finally, the quantification of margins and uncertainties (QMU) methodologies [10] can be refined to develop formal approaches for the flight qualification and certification of various planetary entry systems and technologies through computational analysis and design.

The paper is organized to first describe the basic components of the UQ process in computational models used for planetary entry simulations, which includes information on the classification of uncertainty sources, uncertainty propagation, and utilization of uncertainty analysis results. Section 3 then provides a review of UQ studies applied to planetary entry technologies with the focus on flight mechanics, guidance and navigation simulations, aerothermal modeling and prediction, fluid-structure interaction simulations, and TPS response modeling. Section 4 reviews efficient UQ methods based on stochastic expansions, sparse approximations using compressed sensing, global sensitivity analysis, and dimension reduction. The efficient UQ methods described are then demonstrated on the multidisciplinary analysis of a HIAD configuration in Section 5 with the goal of quantifying the uncertainty in the bondline temperature of flexible TPS during Mars entry. In this section, first a brief description of the problem is given with relevant information on the geometry and trajectory. The following subsections

discuss the computational models, uncertainty sources, and the results for each discipline considered in the UQ study. Finally, the conclusions are given in Section 6.

2. Components of uncertainty quantification process for computational models

In this section, a general uncertainty quantification (UQ) framework is described for analyzing computational models. First, the types of uncertainty in numerical models is discussed, followed by a description of uncertainty propagation through a computational model. The last section briefly discusses how uncertainty-based quantities of interest may be used to aid design efforts and investments.

The UQ process described in the remainder of this section deals primarily with the forward propagation of uncertainty and a frequentist's point of view of statistics. This is to say that the probability distributions of input parameters are assumed to be known and are propagated through the model. The resulting variation in the output is determined based on a population sampling of the inputs.

There is also an entire field of Bayesian-based methods and backward uncertainty propagation that is frequently implemented in UQ studies. With these approaches, assumed distributions of inputs are updated with observed data by using Bayes' theorem. Bayesian-based methods offer a different methodology to achieve, in many cases, the same outcome. The discussion of these methods is beyond the scope of this review and are left to other works [11,12], which the reader is encouraged to investigate.

2.1. Types of uncertainty

A critical step in any uncertainty analysis is identifying the sources of uncertainty and classifying the uncertain parameters appropriately. These parameters may be mathematically represented differently based on the nature of the uncertainty source. Incorrect classification and/or treatment of uncertain parameters can result in widely different ranges of output uncertainty. Particular consideration should be given to identifying and properly classifying the sources of uncertainty in a computational model.

Two main types of uncertainty exist in numerical modeling: aleatory uncertainty and epistemic uncertainty [13]. Aleatory uncertainty is the inherent variation of a physical system. Such variation is due to the random nature of input data and can be mathematically represented by a probability density function if substantial experimental data is available for estimating the distribution type. By definition, these variables are not controllable and are assumed to be as well understood as possible in terms of their uncertain nature. This type of uncertainty is, therefore, sometimes referred to as irreducible uncertainty. In the context of planetary entry modeling, examples of aleatory uncertainty include atmospheric variability, such as density and temperature, vehicle speed, and other quantifiable data that is sufficient to assign a probabilistic distribution.

Epistemic uncertainty in a stochastic problem comes from several potential sources. These include a lack of knowledge or incomplete

information of the behavior of a particular variable. Also, ignorance or negligence with regards to accurate treatment of model input parameters is a source of epistemic uncertainty. Contrary to aleatory uncertainty, epistemic uncertainty is sometimes referred to as reducible uncertainty. An increase in knowledge regarding the physics of a problem, along with accurate modeling, can reduce the amount of this type of uncertainty. Epistemic uncertainty is typically modeled using intervals because the use of probabilistic distributions (even a uniform distribution) can lead to inaccurate predictions in the amount of uncertainty in a system. Upper and lower bounds of these intervals can be drawn from limited experimental data or from expert predictions and judgment [14,15]. In the context of planetary entry modeling, examples of epistemic uncertainty include the transport property model approach used in the CFD simulations and limited data availability in the vehicle structural and thermal properties.

An additional, special case of epistemic uncertainty is numerical error. This uncertainty is common in numerical modeling and is defined as a recognizable deficiency in any phase or activity of modeling and simulations that is not due to lack of knowledge of the physical system. In numerical models such as CFD, an example of this type of uncertainty would be the discretization error in both the temporal and spatial domains that comes from the numerical solution of the partial differential equations that govern the system [15]. This uncertainty can be well understood and controlled through code verification and grid convergence studies.

2.2. Uncertainty propagation

After identifying source of uncertainty, the next step is to propagate the uncertainty "through" the model. In general, this process involves evaluating the computational model at multiple locations within the domain spanned by all of the uncertain parameters to quantify the variation in the quantity of interest. There are multiple approaches that may be taken to accurately quantify this output variation, which may vary depending on the types of uncertainty being considered (e.g., aleatory, epistemic, or both) and the computational burden of the model.

Aleatory uncertainty is propagated through a stochastic model by sampling the probability distribution that describes the variation of each uncertain parameter and then calculating the model output from each random sample set of inputs. Because the inputs are probabilistic in nature, the outputs also carry an associated probability of occurrence. By rigorously sampling within the uncertainty space, the probabilistic realizations can be used to create a discrete cumulative distribution function of the output from the model. Note that a result of the central limit theorem states that the linear combination of independent random variables results in an output distribution that tends to a Gaussian variation.

Epistemic uncertainty can be propagated through a stochastic model in multiple ways. The traditional approach is to rigorously sample the uncertainty space, much like with the aleatory uncertainty. The difference now is that because epistemic uncertainty, by definition, carries no probabilistic meaning, the realizations are viewed simply as possible outcomes, but have the same probability of occurrence as any other output. What is often of interest then is the maximum and minimum possible occurrences. Large sample sizes can be used to approximate these boundaries.

As an alternative, because the primary interest is often in the limits of the uncertain output, optimization on the domain of the epistemic input parameters can be used so seek out the maximum and minimum output values. Eldred and Swiler [16] discuss that this optimization can be performed using a variety of local and global optimization methods. While the optimization approach is generally assumed to provide the most accurate result (assuming that no local extrema is found that is not the interval bound), the computational cost of the optimization may be significant with a very large number of uncertain variables due to

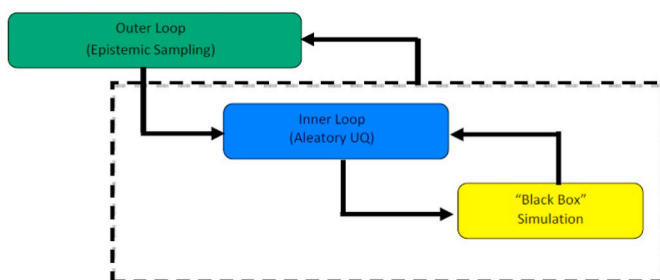


Fig. 4. Schematic of second-order probability.

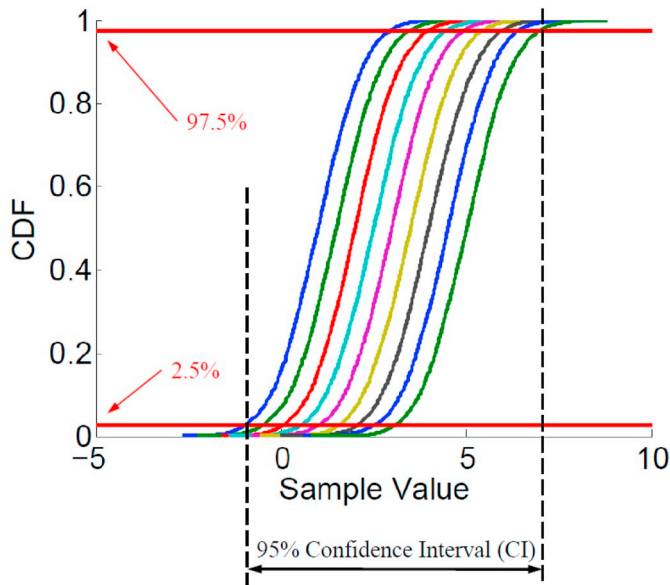


Fig. 5. P-box representation of mixed uncertainty output and confidence interval measurement.

gradient requirements for gradient-based optimization or large sample sizes needed for gradient-free optimization.

Many stochastic problems may contain both epistemic and aleatory uncertainty. The desired approach is to consider the contribution of both types of uncertainty simultaneously by propagating this mixed uncertainty through the stochastic model. This can be done using a procedure known as second-order probability [16]. Second-order probability is a type of double loop sampling, shown in Fig. 4. In the outer loop, a vector of values for the epistemic variables is passed into the inner loop where the stochastic model is sampled for the single epistemic vector and every aleatory vector. Following the repeated process for all epistemic vectors, the total number of samples is the number of epistemic samples times the number of aleatory samples.

Each iteration of the outer loop generates a cumulative distribution function (CDF) based on the aleatory uncertainty analysis in the inner loop. After completion of the process, what remains is a family of CDFs, which, when plotted, gives intervals of the output variable from the model at different probability levels (i.e., a probability or “P-box” representation of mixed uncertainty output). Important information can be taken from P-boxes, including confidence intervals of desired moments. For the case of mixed uncertainty, one approach to obtaining the 95% confidence interval, for example, is to take the upper 97.5% probability level and the lower 2.5% probability level as the interval. The P-box and confidence interval measurement are illustrated in Fig. 5. The boundaries of the P-box typically hold the greatest interest during analysis and design, as will be shown in later sections.

Note that a combination of sampling and optimization can also be implemented. In this approach, the outer loop is first evaluated with a small number of samples to determine robust estimates for the initial values of the epistemic variables used in optimization. The optimization is then performed with these initial starting points for minimizing or maximizing the response at selected probability levels.

Historically, uncertainty propagation has been performed using sampling-based methods, such as Monte Carlo analyses. However, UQ of planetary entry technology modeling requires computational models that are extremely computationally demanding, making sample-based approaches often intractable or infeasible in terms of required time and computational resources. An alternative approach is to use surrogate models to reduce the computational burden. While these methods do have limitations, they have proven to be an efficient and accurate approach to UQ of planetary entry technology modeling. One such

approach is known as stochastic expansions. A review of past studies using this approach is presented in Section 3 and details of the approach as it applies to planetary entry technology modeling is discussed in Section 4.

2.3. Utilization of uncertainty-based results

The last step in a UQ study is to disseminate and use the results. Output quantities obtained from a UQ analysis contain more than just discrete values, but rather describe the statistical behavior of the quantity of interest given the stochastic input parameters. As described in Section 1, this type of information can be used to guide design efforts to accommodate the uncertainty, verification and validation efforts, and systematically guiding resource allocations and investment strategies to better quantify epistemic uncertainty.

In design and analysis, uncertainty-based results can be used to quantify the impact of uncertainty on design performance. Risk and margin assessment can be quantified by statistically providing a scientific and traceable analysis of performance over many traditional margining-based engineering practices. These results can also be used for numerical verification efforts, as well as validation for model improvement when experimental data is available.

To determine where investments would be most effective at better quantifying epistemic uncertainty, sensitivity analyses of the input parameters can be used to rank of the uncertainty sources. With uncertainty in the loop, a global sensitivity analysis provides sensitivity information across the uncertainty domain. This is contrary to traditional, local sensitivities that only provide a gradient at a single point. Examples of commonly-used sensitivity approaches in UQ studies include the global, nonlinear variance decomposition [17], sampling-based global linear regression, and entropy-based sensitivities [18].

3. Review of uncertainty quantification studies applied to planetary entry technologies

In this section, a review of uncertainty quantification studies applied to planetary entry technologies is given. Specific areas covered include flight mechanics, guidance and navigation simulations, aerothermal modeling and prediction, fluid structure interaction simulations, and TPS response modeling. In each of the subsections, the state-of-the-art approach is briefly described, followed by recent advancements in efficient uncertainty quantification approaches that have been applied to the design and analysis of entry vehicles.

3.1. Flight mechanics, guidance and navigation

The problem of studying the evolution of uncertainty in dynamical systems has consistently been of interest in the aerospace community. In particular, the uncertainties in the initial condition and other system parameters for a dynamical system are not known precisely, such as those needed to perform flight dynamics simulations for understanding the location, speed, and path of a hypersonic vehicle during planetary entry. Traditional Monte Carlo approaches have been used extensively in this field for many entry, descent, and landing applications, such as Stardust, Genesis, Mars Phoenix lander, Mars Science Laboratory, and CubeSats [19–22]. Dutta et al. [23] and Kutty and Karlgaard [24] presented methods to assess the uncertainty in parameters of interest using adaptive filtering and reconstruction techniques.

Recently, Prabhakar et al. [25] used an efficient expansion approach for analyzing the uncertainty propagation of a three degree-of-freedom reentry dynamics problem. In order to improve the accuracy and efficiency of the approach for Mars entry dynamics with multiple uncertain variables, a hybrid algorithm using spectral decomposition method was introduced and demonstrated by Jiang and Li [26]. West et al. [10] demonstrated the use of polynomial chaos, which is discussed in more

detail in the next section, in the quantification of margins and uncertainties for a spacecraft reentry problem that consisted of two physics-based systems possessing mixed uncertainty.

3.2. Aerothermal modeling and prediction

Understanding the surface heating and pressure is essential to the design and fabrication of a reliable thermal protection system of an entry vehicle to protect the sensitive underlying structural, mechanical, and electrical components. Early UQ studies have addressed the challenges of present aerothermodynamic simulation capability [27–31]. In particular, a study by Hollis and Prabhu [32] reviewed simulation approaches for convective heating of a Mars entry vehicle. Bose and Wright [33] studied the uncertainty and sensitivities of laminar convective heating for Mars entry using a Monte Carlo approach. Edquist et al. [34] presented a worst-case aerothermodynamic environment prediction for the Mars Science Laboratory mission by setting margins based on a $+3\sigma$ trajectory to account for the uncertainties. Monte Carlo uncertainty and sensitivity studies for Titan atmospheric entry have also been performed by Palmer et al. [35] and Bose et al. [36], which identified key model parameters that have the largest effect on the convective heating. A significant amount of work has also focused on quantifying the uncertainty in the radiative heat transfer prediction, as this component of heating can be a significant source of uncertainty in the aerothermal prediction of planetary entry vehicles [37]. Several papers exist in the literature that present the uncertainty in the radiative heating predictions for lunar-return and hyperbolic Earth entries [38–41], Titan entry [42], and Saturn entry [43] using the traditional Monte Carlo approach.

More recently, efficient techniques in UQ using polynomial chaos have been developed and demonstrated for many entry flow applications. West and Hosder [44] provided an approach to approximate global sensitivities and outlined a multi-step UQ approach. The computational expense for polynomial chaos can also be a problem for large-scale problems with a significant number of uncertain sources. In works by West and Hosder [9] and Doostan and Owadi [45], model problems of large dimension were considered and showed the potential of the method to resolve an accurate polynomial chaos expansion using fewer number of samples. Several other studies exist in the literature that have used polynomial chaos as an efficient means to understand the sensitivities and uncertainties in the aerothermal predictions. Brune et al. [46] investigated the uncertainty in the convective heat flux of a hypersonic inflatable decelerator for Martian entry. Wang et al. [47] and Ju et al. [48] have also performed uncertainty analysis of aeroheating predictions for Mars entry to understand the effects of kinetic rates and geometric parameters. West et al. [49] extended the uncertainty investigation of radiative heating predictions of a hypersonic inflatable aerodynamic decelerator to multiple planetary entries, including Saturn's moon, Titan. West et al. [50] also studied the uncertainty in the afterbody radiative heating prediction of the Stardust capsule.

3.3. Fluid-structure interaction simulations

Shape deformation and aeroelasticity of flexible structures can affect the aerothermodynamic and pressure environments, making these aspects important to the design of deployable entry systems. Pettit [51] provided an early review of UQ in aeroelasticity and emphasized topics in “uncertainty categorization, the propagation of probabilistic inputs to stochastic flutter responses, design methods accounting for aeroelastic reliability, nonprobabilistic and nondeterministic methods for aeroelastic analysis and design, uncertainty in aeroelastic tests, research needs for advancement of UQ in aeroelasticity, and application of UQ to decision making” [52]. Lamorte et al. [53] studied uncertainty propagation using polynomial chaos on a hypersonic panel and demonstrated that deterministic quantification of aeroelastic and aerothermoelastic stability boundaries may be inadequate for hypersonic vehicle analysis.

Brune et al. [54] continued work in analyzing the uncertainty of the hypersonic inflatable aerodynamic decelerator using a coupled fluid-structure interaction approach.

3.4. Thermal protection system response modeling

Another particular application is the design of a TPS for an atmospheric entry vehicle. At sufficiently high velocities during planetary entry, the vehicle will experience significant aerodynamic heating at the surface and can affect the performance of the TPS. Several challenges associated with the development of a credible reliability estimate for TPS sizing have been discussed by Green [55]. Statistics-based material property studies, such as Copeland et al. [56] and Sanson et al. [57], have been conducted to support uncertainty quantification efforts in the modeling and simulation of the material response subject to heating. Monte-Carlo-based methodology was demonstrated for physics-based probabilistic uncertainty analysis of aerothermodynamics and TPS material response modeling of super lightweight and phenolic impregnated carbon ablative heat shields [58,59]. Risk-based approaches and probabilistic modeling techniques have been developed for the design of TPS and planning of ground testing [60,61]. Similar work by Ravishankar et al. [62] focused on uncertainty and reliability estimates for an integrated TPS using Monte Carlo. Many studies also exist in the literature that design the TPS using optimization techniques [63–65]. A worst-case root-sum-square determination of the TPS sizing has also been used in the Crew Exploration Vehicle TPS Margin Management Plan as well as the Mars Science Laboratory margin policy [66–68].

In recent years, flexible thermal protection systems (F-TPS) have become increasingly popular to serve as heat shields for entry vehicles due to advantages in packaging and increased drag surface area. Tobin and Dec introduced a probabilistic design technique using Monte Carlo to determine the material layout needed to keep the bondline temperature below the material limit [69]. To the authors' knowledge, the work by Brune [70] is the only publication found in using polynomial chaos as a means of efficient uncertainty propagation for F-TPS analysis.

4. Uncertainty quantification with stochastic expansions

Since the early 2000's in the age of computing, traditional sampling methods, such as Monte Carlo [71], Quasi-Monte Carlo [72], and importance sampling [73] have been commonly used to perform UQ process described in section 2. However, performing UQ with sampling-based approaches often requires numerous evaluations of the deterministic model to propagate the uncertainty. With the emergence of high-fidelity model techniques to analyze and design complex aerospace systems, efficient approaches, including response surface [74,75] and polynomial chaos based methods [76], have been in increasing demand and promising techniques for quantifying uncertainty in stochastic problems.

More recently, studies began surveying the use of stochastic expansions for efficient UQ in many applications while achieving computational efficiency and accuracy [77,78]. Many works studied efficient approaches to non-intrusive polynomial chaos of mixed uncertainty propagation, including efficient sampling approaches [79,80], point-collocation [14,15], and adjoint-based techniques [81,82]. In these works, sampling-based approaches were still being used, however, the computationally expensive, high-fidelity deterministic model was replaced with a stochastic expansion to greatly reduce the computational burden of uncertainty propagation. Following these developments, advancements in the polynomial chaos techniques were developed to improve computational efficiency while maintaining sufficient accuracy in UQ of many complex hypersonic technology applications [9,44].

4.1. Generalized polynomial chaos

An efficient approach to UQ, polynomial chaos theory, was originally developed by mathematician Norbert Wiener [76] in 1938, when he constructed an orthonormal random basis for expanding a homogeneous chaos using a series of Hermite polynomials. Interestingly, Wiener's proposition was applied by many authors to study problems, such as Burgers equation, parabolic and Navier-Stokes equations, and Brownian motion in early works. The incarnation of the original Wiener polynomial chaos expansions as a numerical method was due to work by Ghanem and Spanos [83], who designed a numerical method for solving elliptical equations with random coefficients. The approach by Ghanem for solving stochastic elliptic equations was further developed and generalized by many other researchers, notably Babuska et al. [84–86], Schwab et al. [87,88], and Keese et al. [89,90]. Xiu and Karniadakis [91] generalized the Hermite-based homogeneous chaos by developing the Wiener-Askey key for parameters of different probability distributions.

Polynomial chaos is a surrogate modeling technique based on the spectral representation of the uncertainty. A response value or random function R can be decomposed into a linear combination of separable deterministic and stochastic components, as shown in Eq. (1).

$$R(\mathbf{x}, \boldsymbol{\xi}) \approx \sum_{i=0}^{N_t-1} \alpha_i(\mathbf{x}) \Psi_i(\boldsymbol{\xi}) \quad (1)$$

Here, α_i is the deterministic component and Ψ_i is the random variable basis functions corresponding to the i^{th} mode. The basis functions, Ψ_i , of each random variable are determined using the Wiener-Askey key [91] and are dependent on the distribution of each random variable. The response, R , is a function of independent, deterministic variables, $\mathbf{x}$, and n independent, standard random variables, $\boldsymbol{\xi}$. Note that this series is, by definition, an infinite series; however, in practice, it is truncated and a discrete sum is taken over a number of output modes. To form a complete basis or a total order expansion, N_t terms are required, which can be computed from Eq. (2) for a polynomial chaos expansion (PCE) of order p and a number of random dimensions or variables, n .

$$N_t = P + 1 = \frac{(n + p)!}{n!p!} \quad (2)$$

4.2. Point-collocation, non-intrusive polynomial chaos

The objective with any PCE method is to determine the expansion coefficients, α_i . To do this, polynomial chaos methods can be implemented using an intrusive or a non-intrusive approach. Intrusive methods require that dependent variables be cast as polynomial chaos expansions. Then, the expansion coefficients that satisfy the input uncertainty distributions must be computed. While an intrusive method may appear straightforward in theory, this process may be time consuming, expensive, and difficult to implement for complex problems [14].

In contrast, the non-intrusive approach can be easily implemented to construct a surrogate model that represents a complex computational simulation, because no modification to the deterministic model is required. The non-intrusive methods require only the response (or sensitivity) [44,81,82] values at selected sample points to approximate the stochastic response surface. While both intrusive and non-intrusive approaches are equivalent and, in general, produce the same results, differences arise due to sampling and regression errors with the non-intrusive approaches and the numerical/solution error with the intrusive approach.

Several methods have been developed for non-intrusive polynomial chaos (NIPC) [77]. Of these, the point-collocation NIPC method has been used extensively in many aerospace simulations and CFD problems [15, 44,80,92] for improved computational efficiency and tractability for high-dimension problems over other spectral projection based

approaches. The point-collocation method starts with replacing a stochastic response or random function with its PCE by using Eq. (1). Then, N_t sample vectors are chosen in the domain spanned by the uncertainty and the deterministic code is evaluated at these points, which is the left hand side of Eq. (1). Following this, a linear system of N_t equations can be formulated and solved for the expansion coefficients of the PCE. This system is shown in Eq. (3).

$$\begin{pmatrix} R(\mathbf{x}, \boldsymbol{\xi}_0) \\ R(\mathbf{x}, \boldsymbol{\xi}_1) \\ \vdots \\ R(\mathbf{x}, \boldsymbol{\xi}_p) \end{pmatrix} = \begin{pmatrix} \Psi_0(\boldsymbol{\xi}_0) & \Psi_1(\boldsymbol{\xi}_0) & \cdots & \Psi_P(\boldsymbol{\xi}_0) \\ \Psi_0(\boldsymbol{\xi}_1) & \Psi_1(\boldsymbol{\xi}_1) & \cdots & \Psi_P(\boldsymbol{\xi}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \Psi_0(\boldsymbol{\xi}_p) & \Psi_1(\boldsymbol{\xi}_p) & \cdots & \Psi_P(\boldsymbol{\xi}_p) \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_p \end{pmatrix} \quad (3)$$

Note that for this linear system, N_t is the minimum number of deterministic samples required to obtain an exact solution. If more samples are available and are linearly independent, the system is considered overdetermined and can be solved using a least squares approach. Hosder et al. [79] determined that the PCE is dependent on the number of collocation points and twice N_t collocation points was sufficient for the stochastic model problems studied.

Polynomial chaos, while powerful and widely applicable, suffers from a “curse of dimensionality” as with any polynomial-based regression method. This means that the number of deterministic model evaluations required to create an accurate surrogate model grows exponentially with the number of random dimensions. For many large-scale, complex problems, such as those found in modeling hypersonic technologies, obtaining even the minimum required number of deterministic model samples may be infeasible or even impossible. The most desirable approach is to obtain an accurate surrogate model with as few deterministic samples as possible to limit the computational cost, even if the minimum number of samples required for a total order expansion is not achievable.

4.3. Sparse approximations using compressed sensing

In 2006, Donoho [93] proposed that if an image or signal is known to be compressible or sparse, a number of measurements much less than the number of pixels may allow for a reconstruction of the full signal or image with only knowledge of what is important. In other words, if a sparse representation is possible, then recovering the sparse approximation may be nearly as accurate, but require far less information to obtain the full representation completely. In terms of polynomial chaos, only a small fraction of the coefficients may carry significant weight in the surrogate model. This would then allow for an assumption that many of the expansion coefficients are zero, making the vector of expansion coefficients sparse. With this assumption, the linear system in Eq. (3) can be regularized allowing for a well-posed solution.

In general, a system of linear equations that has fewer linearly independent equations than unknowns, possesses an infinite number of solutions. The objective is to seek out a solution to the linear system with the fewest number of non-zero coefficients. Using convex relaxation, a solution can be obtained from the L_1 -minimization problem shown in Eq. (4).

$$\min \|\alpha\|_1 \text{ subject to } \|\Psi\alpha - R\|_2 \leq \delta \quad (4)$$

Here, δ is the truncation error associated with the truncation of the series in Eq. (1). If δ is assumed equal to zero, the solution to Eq. (4) can be shown to be unique. In the above formulation, the dimensions of Ψ are $N_s \times N_t$ and the vector R is of length N_s where $N_s < N_t$ for the underdetermined problem. The vector α is of length N_t . Donoho [93], as well as Doostan and Owhadi [45] discuss, in great detail, the theory and formulation of the above method, as well as discussion on stability. Note that convergence on this approach is highly dependent on the sparsity of α meaning that the less sparse the coefficient vector is, the more collocation points, approaching N_t , are needed to recover an accurate sparse approximation.

The optimization problem in Eq. (4) is commonly referred to as Basis

Pursuit Denoising (BPDN) [45,93–95]. This type of problem can be solved using many methods from quadratic programming. Asif et al. [95] developed an efficient least absolute shrinkage and selection operator (LASSO) homotopy optimization routine to find the optimal solution of Eq. (4). While many methods exist for solving the above minimization problem, the homotopy method is used for efficiency, as this method is not significantly affected by the dimensionality of the problem [94].

4.4. Convergence

The optimization and sparse solution recovery approach pose two fundamental issues: (1) how to determine the necessary number of samples, N_s , required to obtain an accurate solution and (2) how to measure the accuracy of the solution. The latter of these assumes that no other means of obtaining the exact solution is possible, thereby relying on the solution obtained from Eq. (4). To reduce the computational cost, the desired approach is to limit the total number of deterministic model evaluations. Addressing both the sample size and the accuracy issues simultaneously requires that there must be a measure of the convergence of the expansion coefficients with increasing sample size.

A naive approach is to first generate an initial sample set of N_t or more random samples and then randomly draw from the sample set, increasing the sample size. Another approach is to use incremental Latin Hypercube sampling (LHS) to gradually grow the sample set, while maintaining a sample structure that covers the space spanned by the uncertain parameters. The idea with both approaches is to start with a small sample structure and evaluate the deterministic model at these points. Then, a first set of PCE coefficients can be obtained using the minimization routine in Eq. (4). This process is then repeated by incrementally expanding the sample structure, which adds more samples to the minimization problem until the convergence of the expansion coefficients is achieved. More sophisticated approaches to choosing samples are the subject of current research in compressed sensing.

After the expansion coefficients are approximated, their convergence should be checked at each iteration. In theory, this could be done by monitoring each individual coefficient. Unfortunately, for large scale problems, there may be thousands of coefficients. Also, because the expansion coefficient vector is known to be sparse, this may not be an accurate approach as the degree of sparseness of the solution vector may decrease with increasing sample size causing radical changes in any convergence error measurement. A logical choice for a convergence metric would be to use output statistics based on the expansion coefficients. Sobol indices [96] can be used to measure the convergence of the solution and are derived via Sobol Decomposition, which is a variance-based global sensitivity analysis method. Note that using the mean to monitor convergence is possible, but may be misleading as the mean often converges rapidly, even though the variance is still converging.

The accuracy of the Sobol indices depends highly on the accuracy of the PCE coefficients, making it an ideal measure of their convergence. Also, because the number of total Sobol indices $S_{T,j}$ (or the total contribution from each variable) is the same as the number of uncertain variables, there are less parameters to track for convergence, as this number will always be less than the number of PCE coefficients. To monitor the convergence of the total Sobol indices with the addition of more samples at each iteration, an absolute error, $S_{e,j}$ can be defined for the j^{th} total Sobol index at iteration i using Eq. (5).

$$S_{e,j} = \| S_{T,i,j} - S_{T,i-1,j} \| \quad (5)$$

Note that measuring the convergence based on this absolute error puts emphasis on the variables that contribute to the output uncertainty more significantly. The errors of each total Sobol index, at each iteration, can then be averaged given a single value for monitoring, which is shown in Eq. (6).

$$\mu_{e,i} = \frac{1}{n} \sum_{j=1}^n S_{e,i,j} \quad (6)$$

Plotting this average error at each iteration would then illustrate the convergence of the PCE coefficients. The objective is to seek out nearly asymptotic convergence, as zero error would likely not be achievable simply due to the randomness of the samples added at each iteration and any numerical inaccuracies that may occur during the analysis of complex models.

4.5. Dimension reduction

For problems with a large number of uncertain variables, the expectation is that many of them will weakly contribute to the overall output uncertainty; however, these parameters may introduce noise to the response surface, which may degrade the surrogate model accuracy. One approach may be to reduce the number of uncertain variables with a small number of deterministic model evaluations by using a sparse approximation and then perform an accurate uncertainty analysis of the problem with the reduced dimensions by using, again, the fully determined or oversampled point-collocation NIPC method.

Using the Sobol index based convergence technique outlined in the previous subsection, the key uncertain variables are implicitly identified while tracking the convergence of the expansion coefficients. West et al. [9,49,50] showed that a relative ranking of the uncertain parameters can be achieved rapidly, with a small sample set for those problem investigated, even if the accuracy of the surrogate model can still be improved by adding additional samples. While the accuracy of the surrogate model may still be in question with a small sample size, the converged ranking of the uncertain parameters will allow for a systematic means of reducing the number of random dimension to a size that may allow for a more accurate surrogate model that is a function of only those uncertain parameters deemed to be of significant importance. Note that there may be multiple levels of dimension reduction to ensure accurate relative ranking of the uncertain parameters as the number of parameters is reduced.

5. Uncertainty quantification applied to the multidisciplinary analysis of a hypersonic inflatable aerodynamic decelerator

In an effort to identify the most viable investment path to transition from current metric-ton class robotic missions to human-scale exploration missions [97], recent work, such as the Entry, Descent, and Landing System Analysis [1,98,99] (EDL-SA), have been tasked to evaluate vehicle concepts and technological investments required to enable human and large-scale payload missions to Mars. Architecture-level assessments of these studies have shown that it is impossible to safely land systems with heritage EDL systems. Multiple deceleration concepts were identified to meet enhanced payload capability. In particular, a HIAD has been considered to be a viable candidate to replace the traditional rigid aeroshell. The inflatable heatshield is packed and stowed to fit within the current launch vehicle size and deployed prior to atmospheric entry, resulting in a heatshield size much larger than the stowed diameter. Therefore, the HIAD provides the advantage of increased payload capability or high-altitude deceleration for pinpoint landing.

NASA's HIAD technology development project recently tested a 6-m stacked torus inflatable article. Multiple aerodynamic wind tunnel and static pressure load tests [2,100,101] were conducted to characterize the inflatable structure response under relevant mission loading. In addition, the HIAD technology development project has undergone extensive aerothermal arc-jet stagnation testing of candidate F-TPS materials and layups to evaluate thermal performance and provide boundary condition and in-depth measurement data for thermal model correlation and validation [69,102]. Accurate prediction of the structural and thermal

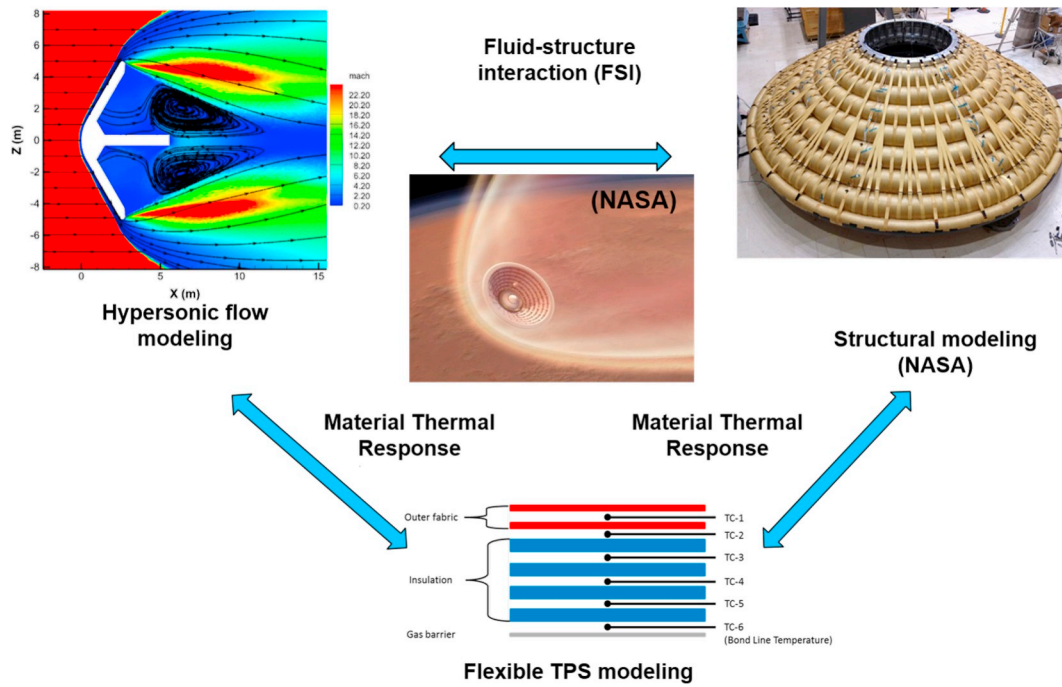


Fig. 6. Multidisciplinary analysis of the HIAD under uncertainty.

response of the HIAD exposed to high aerothermal and mechanical loads presents a challenge in the presence of uncertainties associated with the high-fidelity models used. Therefore, it is important to identify the key factors that impact the performance metrics of each discipline. As an example, the understanding of the aerodynamic heating uncertainty subject to deformations and the flow-field model parameters is important. The aerodynamic heating uncertainty further plays a role in the design of the flexible thermal protection system based on the performance of the bondline temperature, which must be maintained at or below the thermal limits of the impermeable gas barrier and underlying inflatable structure.

In this section, the stochastic expansion based UQ methods described in the previous section will be applied to the multidisciplinary analysis of the HIAD, as shown in Fig. 6. Three disciplines are considered in the modeling of the HIAD: hypersonic flow modeling to obtain the surface heat flux and pressure prediction, structural modeling to obtain the global shape deflections of the aeroshell and how this impacts the surface heat flux and pressure predictions, and the F-TPS thermal modeling to obtain the in-depth temperature response prediction within the material layers. Further details on the reference geometry/components of the HIAD and the nominal trajectory will be described in the next subsection. At the beginning of each subsection that follows, a brief description of the computational models and considered uncertainty sources will be given for each discipline, including the hypersonic flow field (Section 5.2), fluid-structure interaction (Section 5.3) and F-TPS (Section 5.4). Then, the global non-linear sensitivity results will be shown in Sections 5.2 and 5.3 to identify the main contributors in the total variance of the heat flux and pressure that affect the response in the F-TPS. In Section 5.4, the uncertainty and sensitivity results for the bondline temperature will be presented following the model and uncertainty source discussion. Although this section focuses on an example of the HIAD as an application of planetary entry technologies, the approach and processes used in the UQ analysis can be, in general, applied to other engineering systems and disciplines that are not presented here.

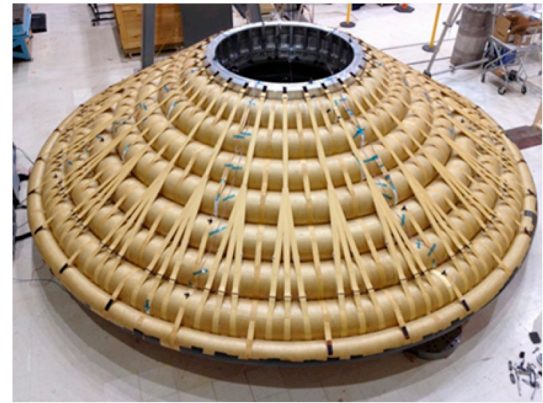


Fig. 7. 6 m test article at NFAC test facility [100,101].

5.1. Nominal trajectory and reference geometry

A 10-m stacked torus configuration is used and scaled from the 6-m article that was tested in the 40-foot x 80-foot test section of the NASA Ames National Full-Scale Aerodynamics Complex (NFAC) facility. The 6-m HIAD (shown in Fig. 7) has a 60-deg sphere-cone stacked-torus design that consists of seven major tori (T1-T7) 14 inches in diameter and an eighth torus at the shoulder 8 inches in diameter (T8). The toroids are modeled as an orthotropic laminate smear with a combination of the Technora bias braid, Kevlar webbing, Urethane membrane, and structural adhesive material properties [100]. In addition, Technora cords are also incorporated as a separate entity to improve the stiffness of the structure and sustain the shape. The inflatable structure described is bonded and attached to a 1.7 m diameter rigid centerbody with a series of Kevlar webbing straps. The Kevlar webbing straps distribute flight loads from torus to torus and to the centerbody. The network of straps include 4000-lb (4-K) load radial and centerbody attachment straps attached to T5 and 3000-lb (3-K) load pairing and chevron straps attached to T6 and T7 [100,101]. Fig. 8 depicts the strap and Technora axial cord components. The properties of the cords and straps are

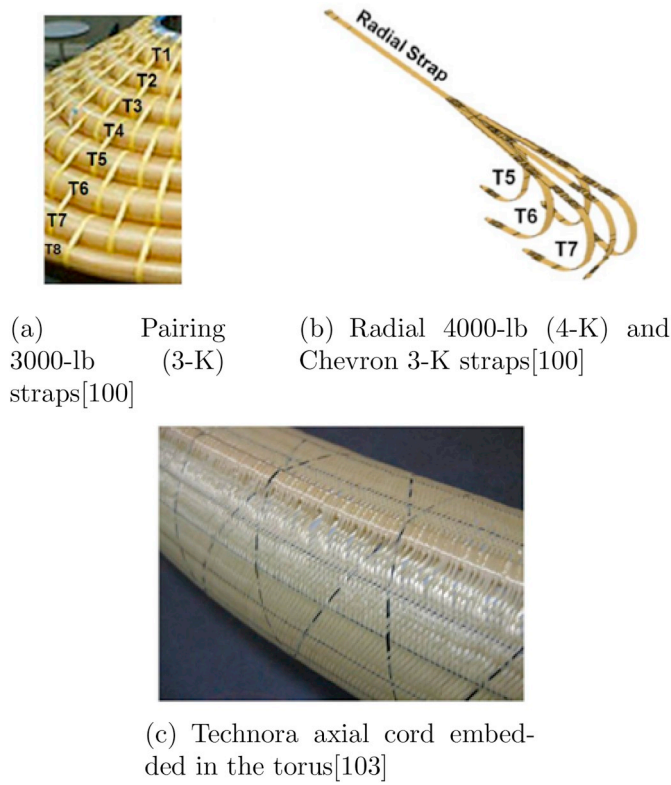


Fig. 8. HIAD Kevlar strap and Technora cord configuration.

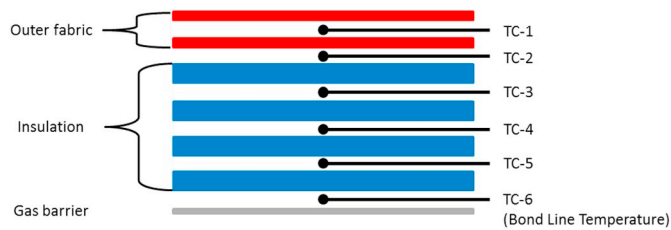


Fig. 9. General TPS layout.

modeled with non-linear elasticity and assumed as isotropic. The torus structure properties are structurally modeled with orthotropic behavior and linear elasticity. The inflation pressure is set to 15 psi [103].

The inflatable structure is covered by the F-TPS from the sphere-cone juncture and around the shoulder. The TPS thermal response analysis, therefore, considers the flank aerodynamic heating and surface pressures obtained from CFD analysis. A one-dimensional F-TPS material layout, shown in Fig. 9, can be used for F-TPS thermal response calculations at each CFD-computed surface location in the flank and shoulder regions. On the right, designated locations are defined in between the material layers. The layout of the F-TPS consists of two outer fabric layers, four insulation layers, and a gas barrier layer, which are labeled on the left in Fig. 9. The outer fabric protects the underlying insulation layers from direct exposure to the convective heat flux and aerodynamic shear forces and from the abrasion and mechanical forces associated with small-volume packaging and deployment. The insulator layers reduce thermal soak back, and the gas barrier prevents hot gas impingement on to the inflatable structure. Of particular interest is the maximum bondline temperature at TC-6, which should not exceed the temperature limit of 400° Celsius (°C). Accurately predicting the bondline temperature ensures a safe and reliable TPS design, given that the most interior layer does not fail. Therefore, this study focuses on predicting the uncertainty in the bondline temperature of the F-TPS subject

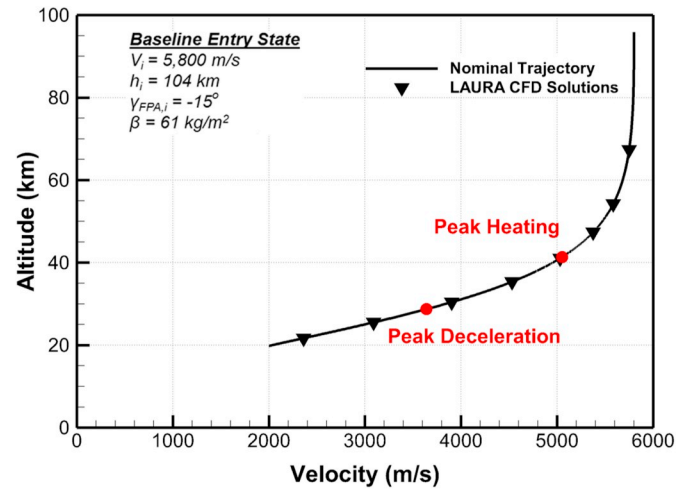


Fig. 10. Ballistic entry profile for EFF-4 reference mission.

Table 1

Trajectory points and corresponding reference conditions for F-TPS uncertainty analysis.

Traj. Point	Time (sec)	AoA (deg)	Yaw (deg)	V_∞ (m/s)	ρ_∞ (kg/m ³)	T_∞ (K)
1	0.0	0.0	0.0	5800.0	2.591e-6	139.0
2	20.0	0.0	0.0	5749.3	3.440e-5	141.9
3	30.0	0.0	0.0	5584.2	1.123e-4	150.3
4	35.6	0.0	0.0	5375.4	2.081e-4	156.2
5	41.2	0.0	0.0	5031.6	3.686e-4	161.1
6	46.8	0.0	0.0	4531.2	6.163e-4	166.9
7	52.4	0.0	0.0	3903.3	9.621e-4	174.6
8	59.2	0.0	0.0	3088.9	1.285e-3	180.7
9	66.0	0.0	0.0	2361.2	2.117e-3	186.9

to uncertainties in the aerodynamic heating and pressure at the surface and the material properties.

An Exploration Feed Forward study [1] investigated an architecture with a 5.8 km/s direct Mars entry. Fig. 10 shows the nominal trajectory corresponding to the EFF4 architecture, derived from a six degree-of-freedom (6-DOF) entry dynamics simulation, with selected points for CFD computations. The selected points are shown to be sufficient in capturing the heat pulse with a total heat load of 1500 J/cm², while meeting continuum flow constraints based on Knudsen number [70]. At 70 s, the HIAD is assumed to separate during transition to a

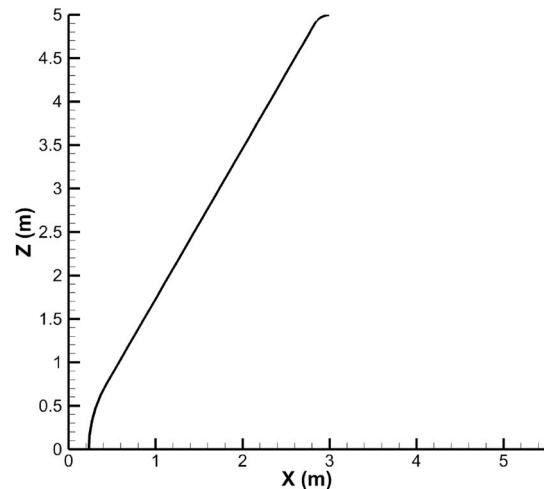


Fig. 11. HIAD surface orientation.

Table 2

Summary of uncertain parameters for the HIAD CFD simulations.

Category	No.	Classification	Uncertainty	Ref.
Chemical Kinetic Reaction Rates	20	Epistemic	Table in Ref.	Johnston et al. [110]
Binary Collision Integrals	36	Epistemic	$\pm 30\%$	Bose et al. [33]
Freestream Velocity	1	Normal	0.5% CoV	Brune et al. [46]
Freestream Density	1	Uniform	$\pm 30\%$	Brune et al. [46]
Freestream Temperature (K)	1	Uniform	$\pm 5\%$	Brune et al. [46]
TOTAL	59			

secondary descent technology (i.e., supersonic retro-propulsion).

As will be shown in the next few subsections, a sensitivity analysis of the convective heating and surface pressure is performed throughout the trajectory at the CFD points with corresponding freestream conditions in Table 1. The freestream composition is assumed to be 96% CO₂ and 4% N₂ by mole. Sensitivities of the deflection and resulting surface heating and pressure from FSI analysis are obtained at the peak heating condition of the trajectory, which is located at point '4'. The uncertainty in the F-TPS response is quantified using the main contributors to the surface heating and pressure variance and the uncertain F-TPS thermal model parameters.

5.2. Hypersonic flow field

For all CFD calculations in Table 1, the Langley Aerothermodynamic Upwind Relaxation Algorithm (LAURA) Navier-Stokes solver [104] from NASA Langley Research Center is used for the high-fidelity numerical solution of the hypersonic flow field over a HIAD, which is initially analyzed as a rigid smooth surface, as shown in Fig. 11. The surface is oriented with the x-axis aligned with the centerline in the direction aft of the nose tip and the z-axis aligned with the radial direction relative to the nose tip. The steady-state flow field is modeled using a two-temperature, thermochemical nonequilibrium model for a Mars 16-species composition: CO₂, CO, N₂, O₂, NO, C, N, O, CN, C₂, C⁺, O⁺, CO⁺, O⁺2, NO⁺, and e⁻. The two-temperature thermochemical nonequilibrium model, applied in this work, is presented by Park et al. [105], with chemical rate modifications based on recent comparisons with shock tube radiation measurements. The chemical rates, with inclusion of these modifications, are presented in a previous study by Brune et al. [46] This analysis conservatively models the boundary layer as fully turbulent with the Cebeci-Smith algebraic turbulence model [106]. The wall of the HIAD is modeled conservatively as super-catalytic, where species recombine at the surface to the freestream composition, for computing the convective heat flux [46].

Fifty nine (59) uncertain variables related to flow field modeling of a HIAD are considered in analyzing the sensitivity of the convective heat flux and surface pressure. Table 2 summarizes the various uncertainty source categories, number of uncertain parameters included in each category, and assigned uncertainty ranges with indicated references. Discussion of model details and references [33,36,107,108] can be found in Brune et al. [46] The uncertain sources include chemical kinetic rates, binary collision interactions for transport property calculations based on Gupta curve fits [109], and freestream conditions.

With a total of 59 uncertain parameters considered as input to the CFD model, the PCEs require a minimum of 1830 CFD evaluations for a second-order polynomial expansion. Using the procedures outlined in Section 4 for the stochastic sparse approximation approach, an initial sample set is generated with 1830 samples using Latin hypercube sampling. A subset of 10 samples are randomly selected at the first iteration to construct a response surface for convective heat flux and surface pressure, and 10 samples are added at each iteration until convergence is

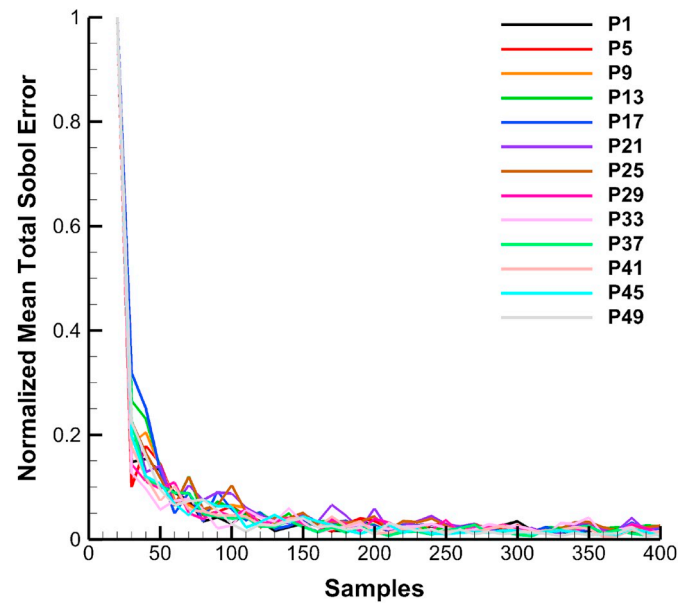


Fig. 12. Convergence of the response surfaces for all LAURA-computed surface points.

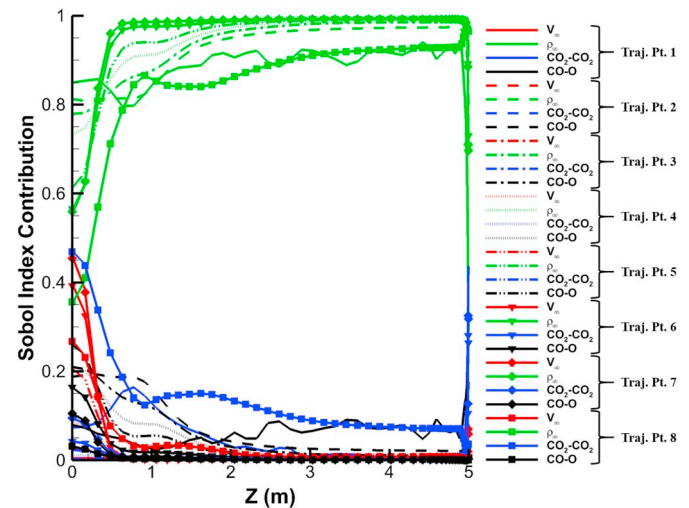


Fig. 13. Sensitivity contributions to the convective heating uncertainty for the entire HIAD surface.

achieved. As an example, Fig. 12 presents the convergence of the mean total Sobol error, normalized by the maximum, for the response surfaces of the convective heat flux that span all 49 computed surface locations on the HIAD. Significant variation of the error is shown for small sample sizes. As the sample size increases, the error drops rapidly to errors of approximately 0.02 at 400 samples. Compared to the 1830 samples required for a second-order expansion using point-collocation, the stochastic expansion approach with sparse approximation is able to recover convergence of the polynomial chaos expansion coefficients (and Sobol indices) using 400 CFD evaluations, resulting in 80% reduction in the computational cost.

Fig. 13 presents the main contributors to approximately 95% of the total variance in the aerodynamic (convective) heating for the entire HIAD surface. The UQ analysis was independently applied at each of the CFD-computed surface locations. The results confirm the four main contributors remain unchanged throughout the trajectory. The free-stream density stands out as the most significant contributor to the convective heating uncertainty in the flank and shoulder regions. The

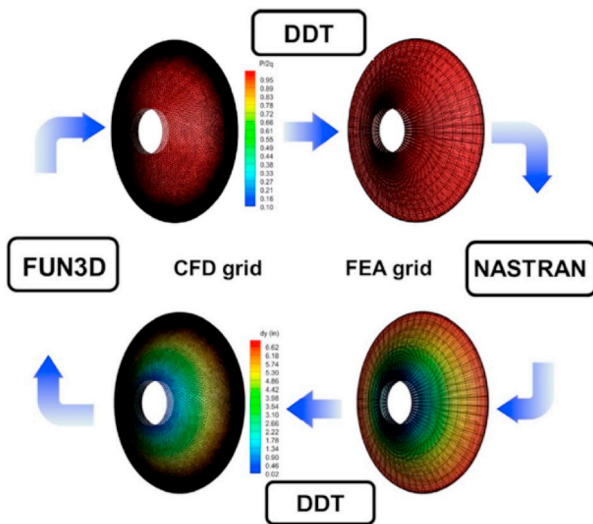


Fig. 14. Coupled FSI process [54].

freestream density contributes to the Reynolds number based on momentum thickness, which also plays a significant role on the turbulent heating augmentation. The freestream velocity, however, has a decreased influence in the convective heating uncertainty downstream of the sphere-cone junction body point. Given a trajectory and atmosphere, these freestream parameters define the undisturbed flow conditions throughout the trajectory prior to the compression shock in front of the HIAD. The CO₂-CO₂ collision pair also significantly impacts the heat flux uncertainty with smaller contribution from the CO₂-O collision pair. These two collision interactions govern the transport property calculations of the flow near the surface of the HIAD. Although not shown here, the surface pressure uncertainty is only impacted by the freestream density.

5.3. FSI response

In the previous subsection, the flow-field sources of uncertainty shown to have the greatest influence on the surface heating and pressure were identified and will now be applied to the FSI response of the HIAD. For FSI calculations at peak heating, the Fully Unstructured Navier-Stokes (FUN3D) flow solver [111] from NASA Langley Research Center is used for the high-fidelity numerical solution of the flow field over a deformable HIAD in the same orientation outlined in the previous subsection. FUN3D uses a three-dimensional node-based, finite-volume discretization approach and includes algorithms for unstructured-grid fluid dynamic simulations of compressible flows with frozen, equilibrium, or nonequilibrium thermo-chemistry. The switch to FUN3D for this particular study was due to the available grid motion algorithms for modeling the HIAD which require rotation, transformation, and deformation. In the previous subsection, the sensitivity analysis of the aerodynamic heating and pressure at the HIAD surface showed that the ionized species have no influence in the surface condition uncertainty. As such, the flow field is computed using the multi-species, two-temperature, thermochemical nonequilibrium model for a Mars 8-species composition: CO₂, CO, N₂, O₂, NO, C, N, and O. The NASTRAN structural solver [112] of MacNeal-Swender Corporation (MSC) is loosely-coupled with the FUN3D flow solver to obtain the structural response (deformation) of the HIAD in the presence of external pressure loads, as shown in Fig. 14. An iterative process is used to communicate between the surface grid of the HIAD used by the flow solver and the full 360-deg structural model with components that resemble the geometry in Fig. 7. The properties of the cords and straps are modeled with non-linear elasticity and assumed as isotropic. The torus structure properties are modeled with orthotropic behaviors and linear elasticity

Table 3

Summary of uncertain parameters for the HIAD FSI simulations.

Uncertain Parameter	Description	Classification	Uncertainty	Ref.
Pressure/Shear Contributions				
V_∞	Freestream Velocity	Normal	0.5% CoV	Brune et al. [54]
ρ_∞	Freestream Density	Uniform	$\pm 30\%$	
A	CO ₂ -CO ₂ Collision Integral	Epistemic	$\pm 30\%$	
Structural Boundary				
P_0	Inflation Pressure	Normal	2% CoV	
Tensile Stiffness				
E_c	Technora Cords	Epistemic	[-50%, +20%]	HIAD Inflatable Structures test data archive
E_{3K}	3-K Kevlar Straps	Epistemic	$\pm 20\%$	
E_{4K}	4-K Kevlar Straps	Epistemic	$\pm 25\%$	
E_L	Torus (Longitudinal)	Epistemic	$\pm 20\%$	
E_T	Torus (Transverse)	Epistemic	$\pm 20\%$	
Possion's Ratio				
ν_c	Technora Cords	Epistemic	$\pm 20\%$	HIAD Inflatable Structures test data archive
ν_{3K}	3-K Kevlar Straps	Epistemic	$\pm 20\%$	
ν_{4K}	4-K Kevlar Straps	Epistemic	$\pm 20\%$	
ν_{LT}	Torus (Uniaxial)	Epistemic	$\pm 15\%$	
Torsional Stiffness				
G_{LT}	Torus (In- plane)	Epistemic	$\pm 25\%$	HIAD Inflatable Structures test data archive
G_{LN}	Torus (Out- plane)	Epistemic	$\pm 25\%$	
G_{TN}	Torus (Out- plane)	Epistemic	$\pm 25\%$	
TOTAL				
	16			

using a nominal internal inflation pressure of 15 psi [103]. Further details on the model assumptions and fluid-structure interaction approach can be found in Brune et al. [54].

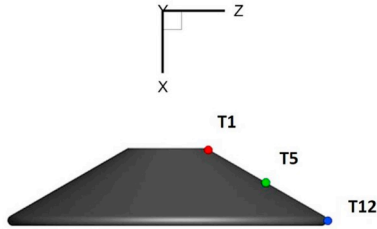
In this study, 16 uncertain input parameters related to flow-field and structural modeling of the HIAD are considered. Table 3 presents the various uncertain source categories and assigned uncertain ranges for the coupled FSI model. In the previous subsection, the first three uncertain sources were shown to contribute approximately 99% of the total uncertainty in the surface conditions of the HIAD with a rigid surface. The remaining structural model uncertainty sources are introduced in this study by surveillance of material property data from experimental load testing and expert opinion from subject matter experts involved with structural modeling of the HIAD [54].

A similar approach to the previous subsection was used to determine the Sobol indices to rank the uncertain variables that contribute to the deflection uncertainty. In Table 4, the uncertain parameters that represent 90% contribution of the deflection uncertainty are listed at eight toroidal locations along the flank (see Fig. 7). The top three contributors are highlighted in bold (first), italics (second), and underline (third). The tensile stiffness of the Technora cord (E_c) is the most significant contributor to the deflection uncertainty over the entire deformable surface. This top contributor has the largest uncertain range compared to other tensile stiffnesses of the torus and straps. The torus tensile stiffnesses (E_L and E_T) and the inflation pressure (P_0) are important towards the nose cap conjunction while the 4-K strap tensile stiffness (E_{4K}) and Possion ratio (ν_{LT}) of the torus become important towards the shoulder. This location-dependent behavior is due to the 4-K straps carrying/distributing loads from the shoulder to the nose cap

Table 4

Uncertain parameter contributions to the HIAD deflection uncertainty.

Parameter	T1	T2	T3	T4	T5	T6	T7	T8
E_c	23.6%	32.9%	66.6%	73.9%	82.5%	85.1%	86.7%	87.2%
E_T	20.2%	16.8%	8.3%	5.9%	2.6%	2.3%	1.3%	1.1%
P_0	14.0%	11.9%	2.5%	1.6%	1.1%	1.0%		
E_L	11.8%	8.6%	2.3%	1.8%				
ν_{LT}	11.8%	11.1%	7.0%	5.3%	3.3%	2.4%	1.6%	1.4%
E_{AK}	9.3%	7.9%	5.0%	4.2%	4.3%	4.0%	4.3%	4.6%

**Fig. 15.** HIAD surface locations for reporting contributions to pressure, shear Stress, and heat flux uncertainty.

centerbody while the inflation pressure and torus structure carry most of the load near the nose cap. The structural stiffnesses dominate the uncertainty in the deflection because of its large magnitude of uncertainty relative to the surface pressure uncertainties, and therefore, segregates the structural uncertainties from the flow-field uncertainties in terms of deflection dependence. The flow-field and freestream uncertainties, however, are expected to be a significant contribution to the uncertainty in the surface conditions in the presence of the deflection. The question that now requires attention is how the deflection uncertainty affects the resulting surface conditions (i.e., surface heat flux and pressure).

Fig. 16 summarizes the contributions of the flow-field and deflection (δ) uncertainty to the surface pressure and heat flux uncertainty at three selected locations on the HIAD surface in Fig. 15. Table 4 showed that the deflection uncertainty was strongly dependent on only structural parameter uncertainties (i.e., stiffnesses of the cords, torus, and straps, and inflation pressure). The deflection uncertainty contributions (δ) reported in Fig. 16 can, therefore, be interpreted as a summation of the Sobol indices of the listed structural parameters in Table 4. In Fig. 16, the freestream density is shown to be the most significant contributor (over 90%) to the surface pressure uncertainty in the flank region of the

HIAD while the deflection uncertainty accounts for no more than 5%. Similar contributions are observed for the convective heat flux uncertainties. Because the contribution of the deflection uncertainty is relatively small compared to the flow-field parameter uncertainties, the deflection was eliminated as a source of uncertainty in the analysis of the F-TPS response in the next subsection.

5.4. F-TPS thermal response

The uncertain sources from CFD and FSI analysis with the greatest influence in the surface heat flux and pressure have been identified in the previous two subsections and will now be applied to the F-TPS thermal response. A one-dimensional physics-based TPS thermal model using COMSOL software is used to obtain the thermal response of the F-TPS in the presence of external pressure and aerodynamic heating loads [113,114]. Heat transfer is modeled within the F-TPS layers to predict the time-dependent in-depth temperatures with given surface pressure and heat flux boundary conditions at the surface. Heat transfer modes considered within the layers of the porous F-TPS materials include solid conduction, radiation, gas conduction, and advection (i.e., convection within porous material). The amount of heat and mass that is transferred through the F-TPS layers is calculated from the solution of the local energy and gas mass conservation equations which are obtained from the theory of the flow of gases through a porous solid [69,114].

Boundary conditions are applied as input at the surface with time dependent pressures and heat flux parameters from LAURA. The flow field is modeled in multi-species, two-temperature, thermo-chemical nonequilibrium for a Mars 8-species composition: CO₂, CO, N₂, O₂, NO, C, N, and O. As already noted in the previous subsection, ionization does not play a significant role in the convective heat flux and pressure predictions. Note that the LAURA-computed heat fluxes are not directly coupled to COMSOL. Instead, LAURA heat flux predictions are obtained at the cold-wall temperature condition, and a correction in the heat flux

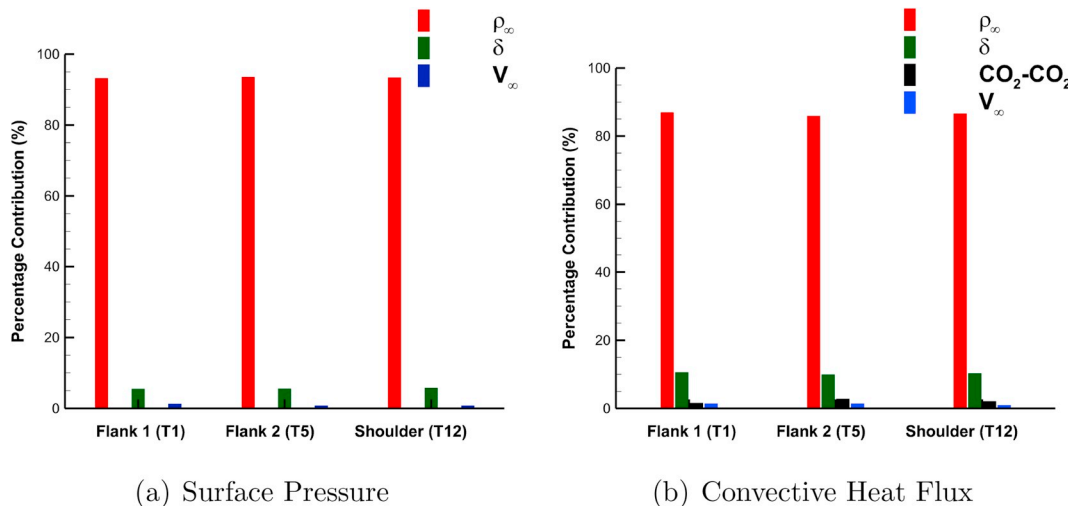
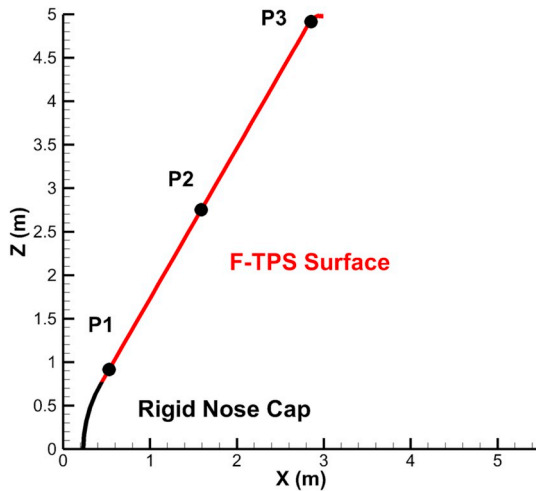
**Fig. 16.** Uncertain parameter contributions to the surface condition uncertainty.

Table 5

Summary of uncertain parameters for the HIAD TPS thermal response simulations.

Component	Parameter	Description	Classification	Uncertainty	Ref.
Convective heating and pressure	ρ_∞	Freestream Density	Uniform	$\pm 30\%$	
	V_∞	Freestream Velocity	Normal	0.5% CoV	[54]
	$A_{\text{CO}_2-\text{CO}_2}$	CO_2 - CO_2 Collision	Epistemic	$\pm 30\%$	[46]
	$A_{\text{CO}_2-\text{O}}$	CO_2 -O Collision	Epistemic	$\pm 30\%$	
Outer Fabric Layers (2)	$K_{s,\text{OF}}$	OF Conductivity	Epistemic	$\pm 30\%$	Same as IN1 uncertainties
	$k_{x,\text{OF}}$	OF Permeability	Epistemic	$\pm 30\%$	
	ε	OF Emissivity	Epistemic	$\pm 10\%$	
	$C_{\text{ps},\text{OF}}$	OF Specific Heat	Epistemic	$\pm 20\%$	
Insulator 1 Layers (3)	$K_{s,\text{IN1}}$	IN1 Conductivity	Epistemic	$\pm 30\%$	Expert Opinion
	$k_{x,\text{IN1}}$	IN1 Permeability	Epistemic	$\pm 30\%$	Expert Opinion
	$C_{\text{ps},\text{IN1}}$	IN1 Specific Heat	Epistemic	$\pm 20\%$	Expert Opinion
	$h_{1,\text{IN1}}$	Layer 1 Thickness	Normal	3% CoV	Expert Opinion
	$h_{2,\text{IN1}}$	Layer 2 Thickness	Normal	3% CoV	Expert Opinion
	$h_{3,\text{IN1}}$	Layer 3 Thickness	Normal	3% CoV	Expert Opinion
Insulator 2 Layers (1)	$K_{s,\text{IN2}}$	IN2 Conductivity	Epistemic	$\pm 30\%$	Expert Opinion
	$k_{x,\text{IN2}}$	IN2 Permeability	Epistemic	$\pm 30\%$	Expert Opinion
	$C_{\text{ps},\text{IN2}}$	IN2 Specific Heat	Epistemic	$\pm 20\%$	Expert Opinion
	Hh_{IN2}	Layer Thickness	Normal	10% CoV	Expert Opinion
	$\rho_{0,\text{IN2}}$	IN2 Virgin Density	Normal	10% CoV	Expert Opinion
	$E_{a,\text{IN2}}$	IN2 Activation Energy	Epistemic	$\pm 20\%$	[116]
Gas Barrier (GB) Layer	$K_{s,\text{GB}}$	GB Conductivity	Epistemic	$\pm 15\%$	Expert Opinion
	$C_{\text{ps},\text{GB}}$	GB Specific Heat	Epistemic	$\pm 5\%$	Expert Opinion
TOTAL	22				

**Fig. 17.** Surface points for F-TPS response analysis for the nominal trajectory.

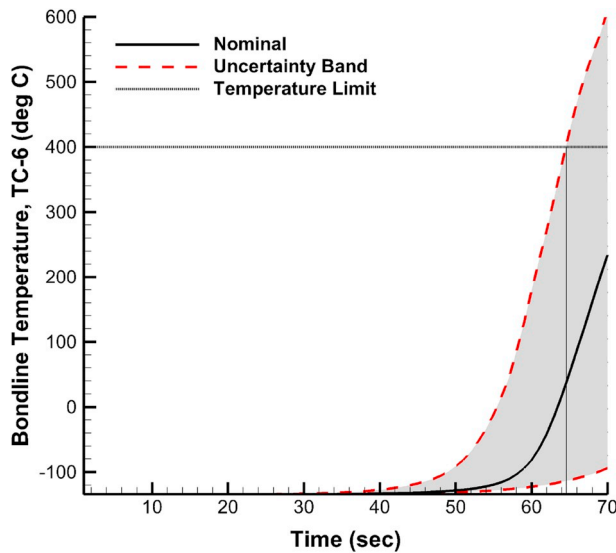
is applied at each time instant based on an update of the wall enthalpy. The wall enthalpy is computed in COMSOL with the freestream species composition (super-catalytic wall condition) and corresponding temperature curve fits [115]. The heat transfer coefficient was shown in a study by Brune et al. [70] to be weakly-dependent on the surface temperature.

A total of 22 uncertain input parameters are considered in this study. Table 5 presents the various uncertainty source categories and assigned uncertainty ranges for the CFD and F-TPS thermal response models used in the prediction of the in-depth temperatures of the F-TPS. The first four uncertainties in Table 5 are the same flow-field sources of uncertainty as those applied in the previous subsections. The remaining uncertain sources listed in Table 5 represent the uncertainty in the F-TPS thermal model. The uncertain ranges are determined by surveillance of material

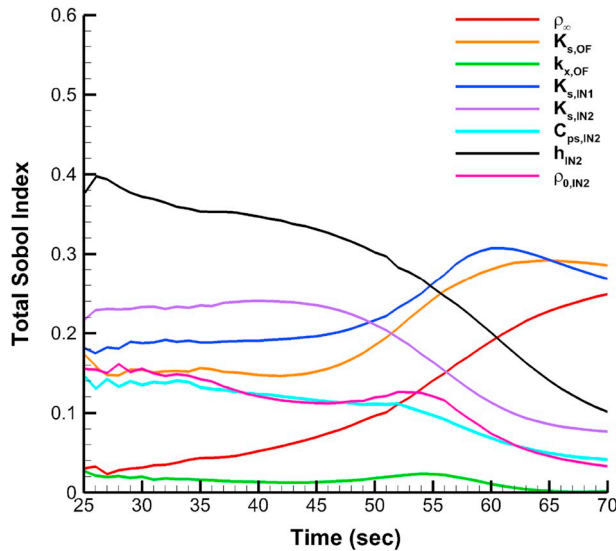
property data from experimental testing and expert opinion from subject matter experts involved with F-TPS modeling of the HIAD. Rossman [116] determined the uncertainty in the activation energy to characterize decomposition of the insulator layer using thermogravimetric analysis.

F-TPS in-depth temperature calculations with COMSOL showed that body point P2 in Fig. 17 is the critical location with maximum bondline temperature along the flank surface [70]. An initial sensitivity study using the procedure outline in Section 4 showed 22 uncertain variables can be reduced to eight based on the Sobol contributions to the total variance in the bondline temperature. A second order PCE surrogate model for the bondline temperature is constructed using 90 evaluations of the F-TPS thermal model, which is twice the minimum number required for a total order expansion ($\text{OSR} = 2$), at 1-s intervals for the 70-s time frame. The uncertainty is then propagated through the surrogate models with 10^6 samples. The uncertainty band of the bondline temperature (95% confidence interval), along with the total Sobol index values for each parameter, for the 70-s time frame are shown in Fig. 18. The uncertainty band is, at a maximum, 125% above and about 50% below the nominal prediction. The greatest uncertainty occurs at 70 s, just prior to HIAD separation for transition to a secondary descent technology. Clearly, the uncertainty in the eight key parameters produces a significant amount of uncertainty in the bondline temperature prediction. At 65 s, the upper bound of the uncertainty band exceeds the 400 °C limit of the bondline temperature. As a result, additional work is needed to determine the thickness requirement, and hence thickness margin, of the F-TPS to ensure the upper bound of the uncertainty band does not exceed the limit state of the bondline temperature. A target reliability level of the bondline temperature is needed to size the F-TPS, which will be investigated in a future study. For the current F-TPS design, the target reliability of the bondline temperature can also be met by reducing the uncertainty in the atmospheric model (i.e., freestream density) and other F-TPS physical parameters with more experiments, measurements, and data.

The Sobol indices in Fig. 18(b) are shown to identify the trends in the



(a) Uncertainty band



(b) Total Sobol indices

Fig. 18. Sensitivities and uncertainty band of bondline temperature with reduced dimension of uncertain variables.

sensitivities as a function of time. At approximately 25 s, the bondline temperature begins to vary from the initial temperature condition of 139 K. Initially, the thickness and thermal conductivity of the insulator 2 layer, near the gas barrier, are the main contributors to the bondline temperature uncertainty prior to an abrupt increase in the temperatures at 50 s. At 50 s, the importance of the insulator 2 layer parameters decrease, and the thermal conductivity of the outer fabric and insulator 1 layer, and the freestream density, become significant. The abrupt increases in the bondline temperature are dictated by the thermal gradients in the outer fabric and insulator 1 layer. In addition, the freestream density uncertainty becomes a major factor further along the trajectory as the HIAD passes through higher densities near peak deceleration.

6. Concluding remarks

The crucial role of computational modeling in planetary entry technology development makes UQ a necessary tool for the assessment of the accuracy of model predictions and improvement of the robustness and reliability of the entry systems. This paper gave a review of UQ studies applied to the modeling and simulation of planetary entry technologies. The review focused on the previous UQ efforts for flight mechanics, guidance and navigation simulations, aerothermal modeling and prediction, fluid structure interaction simulations, and TPS response modeling of entry systems.

This paper also reviewed and described efficient UQ and global sensitivity analysis approaches based on stochastic expansions for high-fidelity simulation of planetary entry technologies. The implementation of stochastic expansions and global sensitivity analysis were demonstrated on the multidisciplinary analysis of a HIAD configuration which included the UQ analysis of shock-layer aerothermodynamics, fluid-structure interaction, and flexible TPS response modeling with the goal of quantifying the uncertainty in the prediction of the TPS bondline temperature for a ballistic Mars entry trajectory. It should be noted that although this work focused on the UQ of a HIAD configuration as an application, the UQ approaches discussed and demonstrated in this paper are general and can be implemented for the uncertainty and sensitivity analysis of other complex engineering systems, such as the modeling of a hypersonic air-breathing flight vehicle.

The sparse approximation stochastic expansion method described and demonstrated in this paper is an effective and efficient uncertainty quantification approach for complex engineering systems with large number of epistemic and aleatory uncertainties including planetary entry technologies; however, certain areas for improvement inherent to the stochastic expansions with global basis functions still exist such as the surrogate modeling of responses with strong discontinuities (e.g., the uncertainty quantification of pressure field in the vicinity of a strong shock wave) and accurate stochastic modeling of time-dependent uncertainty as can be observed for dynamic aeroelastic response modeling of deformable heat shields. In addition, the use of uncertainty and sensitivity information for the reduction of uncertainty in the modeling and simulation of complex aerospace engineering systems require the development of effective verification and validation (V&V) strategies incorporating ground and flight test data with associated uncertainty information. The use of UQ in the simulation-based design of complex engineering systems require the development of effective robust reliability-based design approaches. All these areas warrant further research and improvements to enhance the effectiveness of UQ in the analysis, design, and V&V of planetary entry technologies and other complex engineering systems.

Based on the detailed review and information on UQ of planetary entry systems provided in this paper, the authors strongly believe that UQ will continue to be an essential tool in the development of future entry technologies, which will play a crucial role in the validation and improvement of computational models and increasing the reliability and robustness of the various vehicle systems including TPS.

References

- [1] A.M. Dwyer-Ciancolo, J.L. Davis, W.C. Engelund, D.R. Komar, E.M. Queen, J. A. Samareh, D.W. Way, T.A. Zang, Entry, descent, and landing systems analysis study: phase 2 report on exploration feed-forward systems, Tech. Rep. (Feb. 2011) 1–40. NASA/TM-217055.
- [2] S.J. Hughes, F.M. Cheatwood, R.A. Dillman, H.S. Wright, et al., Hypersonic inflatable aerodynamic decelerator (hiad) technology development overview, AIAA 2011-2524, in: AIAA Aerodynamic Decelerator Systems Technology Conference and Seminar, Dublin, Ireland, 2011, <https://doi.org/10.2514/6.2011-2524>.
- [3] E. Venkatapathy, J. Arnold, K. Fernandez, R. Hamm, D. Kinney, B. Laub, A. Makino, K. McGuire, K. Peterson, D. Prabhu, D. Empey, I. Dupzyk, L. Huynh, P. Hajela, P. Gage, A. Howard, D. Andrews, Adaptive deployable entry and placement technology (adept): a feasibility study for human missions to mars, AIAA 2011-2608, in: 21st AIAA Aerodynamic Decelerator Systems Technology

- Conference and Seminar, Dublin, Ireland, 2011, <https://doi.org/10.2514/6.2011-2608>.
- [4] K.T. Edquist, A.A. Dyakonov, M.J. Wright, C.Y. Tang, Aerothermodynamic environments definition for the mars science laboratory entry capsule, AIAA 2007-1206, in: 45th AIAA Aerospace Sciences Meeting and Exhibit, NV, Reno, 2007, <https://doi.org/10.2514/6.2007-1206>.
 - [5] Hiad site. http://www.nasa.gov/directorates/spacetechnology/game_changing_development/HiAD.
 - [6] Adept video. https://www.youtube.com/watch?v=f_eWC7OZx2E.
 - [7] A.J. Brune, T.K. West, S. Hosder, K.T. Edquist, A review of uncertainty analysis for hypersonic inflatable aerodynamic decelerator design, AIAA 2017-2373, in: 21st AIAA International Space Planes and Hypersonic Technologies Conference, Xiamen, China, 2017, <https://doi.org/10.2514/6.2017-2373>.
 - [8] W.L. Oberkampf, C.J. Roy, Verification and Validation in Scientific Computing, Cambridge University Press, 2010, <https://doi.org/10.1017/CBO9780511760396>.
 - [9] T.K. West IV, S. Hosder, Uncertainty quantification of hypersonic reentry flows with sparse sampling and stochastic expansions, J. Spacecr. Rocket. 52 (2015) 120–133, <https://doi.org/10.2514/1.A32947>.
 - [10] T.K. West IV, S. Hosder, T. Winter, Quantification of margins and uncertainties for integrated spacecraft systems models, J. Spacecr. Rocket. 52 (2015) 450–461, <https://doi.org/10.2514/1.A33067>.
 - [11] P. Lee, Bayesian Statistics, 2004. London, UK.
 - [12] L. Swiler, Bayesian Methods in Engineering Design Problems, Sandia National Laboratories Report SAND2005-3294.
 - [13] W. Oberkampf, J. Helton, K. Sentz, Mathematical Representation of Uncertainty, 2001, <https://doi.org/10.2514/6.2001-1645>. AIAA Paper 2001-1645.
 - [14] S. Hosder, B. Bettis, Uncertainty and sensitivity analysis for reentry flows with inherent and model-form uncertainties, J. Spacecr. Rocket. 49 (2) (2012) 193–206, <https://doi.org/10.2514/1.A32102>.
 - [15] B. Bettis, S. Hosder, T. Winter, Efficient uncertainty quantification in multidisciplinary analysis of a reusable launch vehicle, AIAA 2011-2393, in: 17th AIAA International Space Planes and Hypersonic Systems and Technologies Conference, San Francisco, CA, 2011, <https://doi.org/10.2514/6.2011-2393>.
 - [16] M. Eldred, L. Swiler, Efficient Algorithms for Mixed Aleatory-Epistemic Uncertainty Quantification with Application to Radiation-Hardened Electronics, Sandia National Laboratories Report SAND2009-5805. doi:10.2172/972887.
 - [17] I.M. Sobol, Sensitivity analysis for nonlinear mathematical models, Math. Model. Comput. Exp. 1 (4) (1993) 407–414 (in Russian).
 - [18] T. Zhangchun, L. Zhenzhou, J. Biao, P. Wang, Z. Feng, Entropy-based importance measure for uncertain model inputs, AIAA J. 51 (10) (2013) 2319–2334, <https://doi.org/10.2514/1.J051669>.
 - [19] D.T. Lyons, P.N. Desai, Adventures in parallel processing: entry, descent and landing simulation for the genesis and stardust missions, AAS 05-267, in: AIAA/AAS Astrodynamics Specialist Conference, Lake Tahoe, CA, 2005.
 - [20] J.L. Prince, P.N. Desai, E.M. Queen, M.R. Grover, Entry, descent, and landing operations analysis for the mars phoenix lander, J. Spacecr. Rocket. 48 (5) (2011) 778–783, <https://doi.org/10.2514/1.46563>.
 - [21] D. Way, On the use of a range trigger for the mars science laboratory entry descent and landing, IEEE 1142, in: IEEE Aerospace Conference, Big Sky, MT, 2011, <https://doi.org/10.1109/AERO.2011.5747242>.
 - [22] B. Fritsche, Uncertainty quantification for re-entry survivability prediction, in: T. Sgobba, I. Rongier (Eds.), Space Safety Is No Accident, Springer International Publishing, 2015, pp. 469–478.
 - [23] S. Dutta, R. Braun, C. Karlgaard, Uncertainty quantification for mars entry, descent, and landing reconstruction using adaptive filtering, J. Spacecr. Rocket. 51 (3) (2014) 967–977, <https://doi.org/10.2514/1.A32716>.
 - [24] P. Kutty, C. Karlgaard, Mars science laboratory aerodatabase trajectory reconstruction and uncertainty assessment, AIAA 2014-1094, in: AIAA Atmospheric Flight Mechanics Conference, National Harbor, MD, 2014, <https://doi.org/10.2514/6.2014-1094>.
 - [25] A. Prabhakar, J. Fisher, R. Bhattacharya, Polynomial Chaos-Based analysis of probabilistic uncertainty in hypersonic flight dynamics, J. Guid. Control Dyn. 33 (2010) 222–234, <https://doi.org/10.2514/1.41551>.
 - [26] X. Jiang, Uncertainty quantification for mars atmospheric entry using polynomial chaos and spectral decomposition, AIAA 2018-1317, in: AIAA Guidance, Navigation, and Control Conference, Kissimmee, FL, 2018, 2018, <https://doi.org/10.2514/6.2018-1317>.
 - [27] R. Stogner, P. Bauman, K. Schulz, R. Upadhyay, A. Maurente, Uncertainty and parameter sensitivity in multiphysics reentry flows, AIAA 2011-764, in: 49th AIAA Aerospace Sciences Meeting, Orlando, FL, 2011, <https://doi.org/10.2514/6.2011-764>.
 - [28] K. Miki, M. Panesi, E. Prudencio, S. Prudhomme, Probabilistic models and uncertainty quantification for the ionization reaction rate of atomic nitrogen, J. Comput. Phys. 231 (9) (2012) 3871–3886, <https://doi.org/10.1016/j.jcp.2012.01.005>.
 - [29] D. Bose, J. Brown, D. Prabhu, P. Gnoffo, C.O. Johnston, B. Hollis, Uncertainty assessment of hypersonic aerothermodynamics prediction capability, J. Spacecr. Rocket. 50 (1) (2013) 12–18, <https://doi.org/10.2514/1.A32268>.
 - [30] N. Villedieu, J. Cappaert, J. Garcia Galache, T. Magin, Uncertainty quantification for characterization of high enthalpy facilities, Prog. Flight Phys. 5 (2013) 519–532, <https://doi.org/10.1051/eucass/201305519>.
 - [31] J. Tryoen, P. Congedo, R. Abgrall, N. Villedieu, T. Magin, Bayesian-based method with metamodels for rebuilding freestream conditions in atmospheric entry flows, AIAA J. 52 (10) (2014) 2190–2197, <https://doi.org/10.2514/1.J052831>.
 - [32] B.R. Hollis, D.K. Prabhu, Assessment of laminar, convective aeroheating prediction uncertainties for mars entry vehicles, J. Spacecr. Rocket. 50 (1) (2013) 56–68, <https://doi.org/10.2514/1.A32257>.
 - [33] D. Bose, M. Wright, Uncertainty analysis of laminar aeroheating predictions for mars entries, J. Thermophys. Heat Transf. 20 (4) (2006) 652–662, <https://doi.org/10.2514/1.20993>.
 - [34] K.T. Edquist, A.A. Dyakonov, M.J. Wright, C.Y. Tang, Aerothermodynamic design of the mars science laboratory heatshield, AIAA 2009-4075, in: 41st AIAA Thermophysics Conference, San Antonio, TX, 2007, <https://doi.org/10.2514/6.2009-4075>.
 - [35] G. Palmer, J. Olejniczak, M. Wright, Uncertainty analysis of aeroheating predictions for titan entries, aiaa 2006-0388, in: 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, 2006, <https://doi.org/10.2514/6.2006-388>.
 - [36] D. Bose, M. Wright, T. Gökçen, Uncertainty and sensitivity analysis of thermochemical modeling for titan atmospheric entry, AIAA 2004-2455, in: 37th AIAA Thermophysics Conference, Portland, OR, 2004, <https://doi.org/10.2514/6.2004-2455>.
 - [37] D. Bose, E. McCorkle, C. Thompson, D. Bogdanoff, D. Prabhu, G. Allen, J. Grunstead, Analysis and model validation of shock layer radiation in air, AIAA 2008-1246, in: 46th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, 2008, <https://doi.org/10.2514/6.2008-1246>.
 - [38] C.O. Johnston, B. Kleb, Uncertainty analysis of air radiation for lunar-return shock layers, J. Spacecr. Rocket. 49 (3) (2012) 425–434, <https://doi.org/10.2514/1.A32161>.
 - [39] C.O. Johnston, A. Mazaheri, P. Gnoffo, B. Kleb, D. Bose, Radiative heating uncertainty for hyperbolic earth entry, part 1: flight simulation modeling and uncertainty, J. Spacecr. Rocket. 50 (1) (2013) 19–38, <https://doi.org/10.2514/1.A32254>.
 - [40] C.O. Johnston, K. Sutton, D. Prabhu, D. Bose, Radiative heating uncertainty for hyperbolic earth entry, part 2: comparisons with 1960s-era shock-tube measurements, J. Spacecr. Rocket. 50 (1) (2013) 39–47, <https://doi.org/10.2514/1.A32483>.
 - [41] A. Brandis, C.O. Johnston, B. Cruden, D. Prabhu, D. Bose, Uncertainty analysis and validation of radiation measurements for earth reentry, J. Thermophys. Heat Transf. 29 (2) (2015) 209–221, <https://doi.org/10.2514/1.T4000>.
 - [42] M.T. Ghaffari, G. Iaccarino, Uncertainty quantification of radiative heat flux modeling for titan atmospheric entry, AIAA 2010-239, in: 8th AIAA Aerospace Sciences Meeting, Orlando, FL, 2010, <https://doi.org/10.2514/6.2010-239>.
 - [43] G. Palmer, D. Prabhu, B. Cruden, Aeroheating uncertainties in uranus and saturn entries by the Monte Carlo method, J. Spacecr. Rocket. 51 (3) (2014) 801–814, <https://doi.org/10.2514/1.A32768>.
 - [44] T.K. West IV, S. Hosder, C.O. Johnston, Multistep uncertainty quantification approach applied to hypersonic reentry flows, J. Spacecr. Rocket. 51 (2014) 296–310, <https://doi.org/10.2514/1.A32592>.
 - [45] A. Doostan, H. Owahdi, A non-adapted sparse approximation of pdes with stochastic inputs, J. Comput. Phys. 230 (8) (2011) 3015–3034, <https://doi.org/10.1016/j.jcp.2011.01.002>.
 - [46] A.J. Brune, T. West, S. Hosder, K.T. Edquist, Uncertainty analysis of mars entry flows over hypersonic inflatable aerodynamic decelerators, J. Spacecr. Rocket. 52 (3) (2015) 776–788, <https://doi.org/10.2514/1.A33131>.
 - [47] X. Wang, C. Yan, S. Ju, Y. Zheng, J. Yu, Uncertainty analysis of laminar and turbulent aeroheating predictions for mars entry, Int. J. Heat Mass Transf. 112 (2017) 533–543, <https://doi.org/10.1016/j.ijheatmasstransfer.2017.04.126>.
 - [48] S. Ju, C. Yan, X. Wang, Y. Qin, Z. Ye, Sensitivity analysis of geometric parameters upon the aerothermodynamic performances of mars entry vehicle, Int. J. Heat Mass Transf. 120 (2018) 597–607, <https://doi.org/10.1016/j.ijheatmasstransfer.2017.12.073>.
 - [49] T.K. West IV, A.J. Brune, S. Hosder, C.O. Johnston, Uncertainty analysis of radiative heating predictions for titan entry, J. Thermophys. Heat Transf. 30 (2016) 438–451, <https://doi.org/10.2514/1.T4620>.
 - [50] T.K. West IV, C.O. Johnston, S. Hosder, Uncertainty and sensitivity analysis of afterbody radiative heating for earth entry, J. Thermophys. Heat Transf. 31 (2017) 294–306, <https://doi.org/10.2514/1.T4948>.
 - [51] C.L. Pettit, Uncertainty quantification in aeroelasticity: recent results and research challenges, J. Aircr. 41 (2004) 1217–1229, <https://doi.org/10.2514/1.3961>.
 - [52] P. Beran, B. Stanford, C. Schrock, Uncertainty quantification in aeroelasticity, Annu. Rev. Fluid Mech. 49 (2017) 361–386, <https://doi.org/10.1146/annurev-fluid-122414-034441>.
 - [53] N. Lamorte, P.P. Friedmann, B. Glaz, Uncertainty propagation in hypersonic aerothermoelastic analysis, J. Aircr. 51 (2014) 192–203, <https://doi.org/10.2514/1.C032233>.
 - [54] A.J. Brune, S. Hosder, K.T. Edquist, Uncertainty analysis of fluid-structure interaction of a deformable hypersonic inflatable aerodynamic decelerator, J. Spacecr. Rocket. 53 (4) (2015) 654–668, <https://doi.org/10.2514/1.A33532>.
 - [55] L. Green, The challenges of credible thermal protection system reliability quantification, IPPW 10-77, in: International Planetary Probe Workshop, San Jose, CA, 2013.
 - [56] S. Copeland, M. Mahzari, I. Cozmuta, J. Alonso, A statistics-based material property analysis to support ablation simulation uq efforts, AIAA 2012-1763, in: AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference, Honolulu, HI, 2012, <https://doi.org/10.2514/6.2012-1763>.
 - [57] F. Sanson, F. Panerai, E. Magin, P. Marco Congedo, Robust reconstruction of the catalytic properties of thermal protection materials from sparse high-enthalpy facility experimental data, Exp. Therm. Fluid Sci. 96. doi:10.1016/j.expthermflusci.2018.03.028.

- [58] M. Wright, D. Bose, Y. Chen, Probabilistic modeling of aerothermal and thermal protection material response uncertainties, *AIAA J.* 45 (2) (2007) 399–410, <https://doi.org/10.2514/1.26018>.
- [59] S. Sepka, M. Wright, Monte Carlo approach to fiat uncertainties with applications for mars science laboratory, *J. Thermophys. Heat Transf.* 25 (4) (2011) 516–522, <https://doi.org/10.2514/1.49804>.
- [60] M. Wright, J. Grinstead, A Risk-Based Approach for Aerothermal/tps Analysis and Testing, Tech. rep., RTO-EN-AVT-142.
- [61] D. Bose, M. Wright, N. Mansour, The Use of Probabilistic Modeling for Reentry Thermal Protection System Design, Tech. rep., RTO-MP-AVT-147.
- [62] B. Ravishanker, R. Hafta, B. Sankar, Uncertainty analysis of integrated thermal protection system with rigid insulation bars, *AIAA 2011-1767*, in: *AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference*, Denver, CO, 2011, <https://doi.org/10.2514/6.2011-1767>.
- [63] N. Vlahopoulos, J. He, Designing the thermal protection system of an apollo type vehicle under uncertainty, Tech. Rep. (2009), <https://doi.org/10.4271/2009-01-0564>. SAE 2009-01-0564.
- [64] L. Hou, Y. Cai, J. Li, Evidence-Based Multi-disciplinary robust optimization for mars microentry probe design, in: M. Emmerich, A. Deutz, O. Schütze, P. Legrand, E. Tantar, A. Tantar (Eds.), *EVOLVE – A Bridge between Probability, Set Oriented Numerics and Evolutionary Computation VII*, Springer International Publishing, 2017, pp. 135–155, https://doi.org/10.1007/978-3-319-49325-1_7.
- [65] A. Nenarolomov, Optimal design of multi-layer thermal protection and analysis of uncertainties, in: A. Wilson (Ed.), *Hot Structures and Thermal Protection Systems for Space Vehicles*, Proceedings of the 4th European Workshop, 2003, p. 267.
- [66] M. Wright, Cev Thermal Protection System (Tps) Margin Management Plan, Nov. 2006. Tech. rep., NASA/C-TPSA-A-DOC-7005.
- [67] M. Wright, R. Beck, K. Edquist, D. Driver, S. Sepka, E. Slimko, W. Willcockson, Sizing and margins assessment of mars science laboratory aeroshell thermal protection system, *J. Spacecr. Rocket.* 51 (4) (2014) 1125–1138, <https://doi.org/10.2514/1.A32579>.
- [68] I. Cozmuta, M. Wright, B. Laub, W. Willcockson, Defining ablative thermal protection system margins for planetary entry vehicles, *AIAA 2011-3757*, in: 42nd AIAA Thermophysics Conference, Honolulu, HI, 2011, <https://doi.org/10.2514/6.2011-3757>.
- [69] S.A. Tobin, J.A. Dec, A probabilistic sizing demonstration of a flexible thermal protection system for a hypersonic inflatable aerodynamic decelerator, *AIAA 2015-1895*, in: *AIAA SciTech Conference*, Kissimmee, FL, 2015, <https://doi.org/10.2514/6.2015-1895>.
- [70] A.J. Brune, S. Hosder, K.T. Edquist, S.A. Tobin, Thermal protection system response uncertainty of a hypersonic inflatable aerodynamic decelerator, *J. Spacecr. Rocket.* 54 (1) (2016) 141–154, <https://doi.org/10.2514/1.A33732>.
- [71] R.E. Caflisch, Monte Carlo and quasi-Monte Carlo methods, *Acta Numer.* 7 (1998) 1–49, <https://doi.org/10.1017/S0962492900002804>.
- [72] H. Niederreiter, Quasi-Monte Carlo methods and pseudo-random numbers, *Bull. Am. Math. Soc.* 84 (6). doi:10.1090/S0002-9904-1978-14532-7.
- [73] J. Bucklew, *Introduction to Rare Event Simulation*, Springer, 2004, <https://doi.org/10.1007/978-1-4757-4078-3>.
- [74] R.H. Myers, D.C. Montgomery, C.M. Anderson-Cook, *Response Surface Methodology: Process and Product Optimization Using Designed Experiments*, fourth ed., Wiley, 2016.
- [75] T. Hastie, R. Tibshirani, J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, second ed., Springer, 2009 <https://doi.org/10.1007/978-0-387-84858-7>.
- [76] N. Wiener, The homogenous chaos, *Am. J. Math.* 60 (4) (1938) 897–936, <https://doi.org/10.2307/2371268>.
- [77] M.S. Eldred, Recent advances in non-intrusive polynomial chaos and stochastic collocation methods for uncertainty analysis and design, *AIAA 2009-2274*, in: 50th AIAA/ASME/ASCE/AHS/ASC Structures, Palm Springs, CA, 2009, <https://doi.org/10.2514/6.2009-2274>.
- [78] H.N. Najm, Uncertainty quantification and polynomial chaos techniques in computational fluid dynamics, *Annu. Rev. Fluid Mech.* 41 (2009) 35–52, <https://doi.org/10.1146/annurev.fluid.010908.165248>.
- [79] S. Hosder, R.W. Walters, M. Balch, Efficient sampling for non-intrusive polynomial chaos applications with multiple uncertain input variables, *AIAA 2007-1939*, in: 48th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference, Honolulu, HI, 2007, <https://doi.org/10.2514/6.2007-1939>.
- [80] S. Hosder, R.W. Walters, M. Balch, Point-collocation nonintrusive polynomial chaos method for stochastic computational fluid dynamics, *AIAA J.* 48 (12) (2010) 2721–2730, <https://doi.org/10.2514/1.39389>.
- [81] B. Lockwood, D. Mavriplis, Gradient-Based methods for uncertainty quantification in hypersonic flows, *Comput. Fluids J.* 85 (2013) 27–38, <https://doi.org/10.1016/j.compfluid.2012.09.003>.
- [82] O. Roderick, M. Anitescu, P. Fischer, Polynomial regression approaches using derivative information for uncertainty quantification, *Nucl. Sci. Eng.* 164 (2) (2010) 122–139, <https://doi.org/10.13182/NSE08-79>.
- [83] R.G. Ghanem, P.D. Spanos, *Stochastic Finite Elements: A Spectral Approach*, Springer-Verlag, New York, 1991, <https://doi.org/10.1007/978-1-4612-3094-6>.
- [84] I. Babuska, P. Chatziantelidis, On solving elliptic stochastic partial differential equations, *J. Comput. Methods Appl. Mech. Eng.* 191 (37–38) (2002) 4093–4122, [https://doi.org/10.1016/S0045-7825\(02\)00354-7](https://doi.org/10.1016/S0045-7825(02)00354-7).
- [85] I. Babuska, R. Tempone, G.E. Zouraris, Galerkin finite element approximations of stochastic elliptic partial differential equations, *SIAM J. Numer. Anal.* 42 (2) (2004) 800–825, <https://doi.org/10.1137/S0036142902418680>.
- [86] I. Babuska, R. Tempone, G.E. Zouraris, Solving elliptic boundary value problems with uncertain coefficients by the finite element method: the stochastic formulation, *J. Comput. Methods Appl. Mech. Eng.* 194 (12–16) (2005) 1251–1294, <https://doi.org/10.1016/j.cma.2004.02.026>.
- [87] C. Swab, R.A. Todor, Sparse finite elements for elliptic problems with stochastic loading, *J. Numer. Math.* 95 (4) (2003) 707–734, <https://doi.org/10.1007/s00211-003-0455-z>.
- [88] P. Faruenfelder, C. Swab, R.A. Todor, Finite elements for elliptic problems with stochastic coefficients, *J. Comput. Methods Appl. Mech. Eng.* 194 (2–5) (2005) 205–228, <https://doi.org/10.1016/j.cma.2004.04.008>.
- [89] A. Keese, A Review of Recent Developments in the Numerical Solution of Stochastic Partial Differential Equations (Stochastic Finite Elements), Oct. 2003. Tech. rep., Institute of Scientific Computing, Technical University Braunschweig.
- [90] H.G. Matthies, A. Keese, Galerkin methods for linear and nonlinear elliptic stochastic partial differential equations, *J. Comput. Methods Appl. Mech. Eng.* 194 (12–16) (2005) 1295–1331, <https://doi.org/10.1016/j.cma.2004.05.027>.
- [91] D. Xiu, G.E. Karniadakis, The wiener–askey polynomial chaos for stochastic differential equations, *SIAM J. Sci. Comput.* 24 (2) (2002) 619–644, <https://doi.org/10.1137/S1064827501387826>.
- [92] D. Han, S. Hosder, Inherent and model-form uncertainty analysis for cfd simulation of synthetic jet actuators, *AIAA 2012-0082*, in: 48th AIAA Aerospace Sciences Meeting, Nashville, TN, 2012, <https://doi.org/10.2514/6.2012-82>.
- [93] D.L. Donoho, Compressed sensing, *IEEE Trans. Inf. Theory* 52 (4) (2006) 1289–1306, <https://doi.org/10.1109/TIT.2006.871582>.
- [94] A. Yang, A. Ganesh, S. Sastry, Y. Ma, Fast l1-minimization algorithms and an application in robust face recognition: a review, Tech. Rep. (Feb. 2010) 1–4. Tech. Rep. UCB/EECS-2010-13.
- [95] M.S. Asif, J. Romberg, Fast and accurate algorithms for re-weighted l1-norm minimization, *IEEE Trans. Signal Process.* 61 (23) (2013) 4905–4916, <https://doi.org/10.1109/TSP.2013.2279362>.
- [96] B. Sudret, Global sensitivity analysis using polynomial chaos expansion, *Reliab. Eng. Syst. Saf.* 93 (7) (2008) 964–979, <https://doi.org/10.1016/j.res.2007.04.002>.
- [97] B.G. Drake, Human exploration of mars design reference architecture 5.0, Tech. Rep. (2009), <https://doi.org/10.1109/AERO.2010.5446736>. NASA-SP-2009-566.
- [98] T.A. Yang, A.M. Dwyer-Ciancolo, D.J. Kinney, A.R. Howard, G.T. Chen, M. C. Ivanov, R.R. Sostaric, M.C. Westhelle, Overview of the nasa entry, descent, and landing systems analysis studies for large robotic-class missions, *AIAA 2010-8649*, in: *AIAA SPACE 2010 Conference and Exhibition*, Anaheim, CA, 2010, <https://doi.org/10.2514/6.2011-7294>.
- [99] A.M. Dwyer-Ciancolo, J.L. Davis, J.D. Shidner, R.W. Powell, Entry, descent, and landing systems analysis: exploration class simulation overview and results, *AIAA 2010-7970*, in: *AIAA/AAS Astrodynamic Specialist Conference*, Toronto, Ontario Canada, 2010, <https://doi.org/10.2514/6.2010-7970>.
- [100] G. Swanson, A.M. Cassell, R.K. Johnson, S.J. Hughes, A.M. Calomino, F. M. Cheatwood, Structural strap tension measurements of a 6-meter hypersonic inflatable aerodynamic decelerator under static and dynamic loading, *AIAA 2013-1287*, in: 22nd AIAA Aerodynamic Decelerator Systems Technology Conference, Daytona Beach, FL, 2013, <https://doi.org/10.2514/6.2013-1287>.
- [101] A.M. Cassell, G.T. Swanson, B.T. Quach, L.K. Kushner, et al., Design and execution of the hypersonic inflatable aerodynamic decelerator large-article wind tunnel experiment, *AIAA 2013-1304*, in: *AIAA Aerodynamic Decelerator Systems Conference*, Daytona Beach, FL, 2013, <https://doi.org/10.2514/6.2013-1304>.
- [102] W.E. Bruce III, M.N. J. P.G. Furelmann, P.M. Siemers III, J.A. Del Corso, S. J. Hughes, S.A. Tobin, M.P. Kardell, Aerothermal ground testing of flexible thermal protection systems for hypersonic inflatable aerodynamic decelerators, in: 9th International Planetary Probe Workshop, Toulouse, France, 2012.
- [103] L. Lichodziejewski, B. Tutt, J. Jurewicz, R. Dillman, K. Johnson, A.M. Cassell, Ground and flight testing of a stacked tori hypersonic inflatable aerodynamic decelerator configuration, *AIAA 2013-1864*, in: 54th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference, Boston, MA, 2013, <https://doi.org/10.2514/6.2013-1864>.
- [104] A. Mazaheri, P.A. Gnoffo, C.O. Johnston, B. Kleb, *Laura Users Manual: 5.5-65135*, Tech. Rep. Feb. 2012. NASA/TM-2011-217800.
- [105] C. Park, J.T. Howe, R.L. Jaffe, G.V. Candler, Review of chemical-kinetic problems for future nasa missions, ii: mars entries, *J. Thermophys. Heat Transf.* 8 (1994) 9–23.
- [106] A.M. Smith, T. Cebeci, Numerical solution of the turbulent boundary layer equations, Tech. Rep. (1967) 1–142. Tech. Rep. Douglas Aircraft Division DAC-33735.
- [107] M.J. Wright, H.H. Hwang, D.W. Schwenke, Recommended collision integrals for transport property computations part ii: mars and venus entries, *AIAA J.* 45 (1) (2007) 281–288, <https://doi.org/10.2514/1.24523>.
- [108] G. Palmer, Uncertainty analysis of cev leo and lunar return entries, *aiaa 2007-4253*, in: 39th AIAA Thermophysics Conference, Miami, FL, 2007, <https://doi.org/10.2514/6.2007-4253>.
- [109] R.N. Gupta, J.M. Yos, R.A. Thompson, K.P. Lee, A review of reaction rates and thermodynamic and transport properties for an 11-species air model for chemical and thermal nonequilibrium calculations to 30000 k, Tech. Rep. (Aug. 1990) 1–86. NASA/RP-1232.
- [110] C.O. Johnston, A.M. Brandis, K. Sutton, Shock layer radiation modeling and uncertainty for mars entry, *AIAA 2012-2866*, in: 43rd AIAA Thermophysics Conference, New Orleans, LA, 2012, <https://doi.org/10.2514/6.2012-2866>.
- [111] R. T. Biedron, J. M. Derlaga, P. A. Gnoffo, D. P. Hammond, et al., *Fun3d Manual: 12.4*, Tech. Rep., NASA/TM-2014-218179 (Mar. 2014).

- [112] MSC, Msc Nastran 2013.1.1 Release Guide, Mar. 2014. Tech. rep., MSC/DOC-10524.
- [113] COMSOL, Comsol Multiphysics User's Guide, v5.0, Oct. 2014. Tech. rep., COMSOL/CM-010004.
- [114] R.M. Sullivan, E.H. Baker, J.A. Dec, Analysis methods for heat transfer in flexible thermal protection systems for hypersonic inflatable aerodynamic decelerators, Tech. Rep. (Apr. 2014) 1–35. NASA/TM-2014-218311.
- [115] S. Gordon, B. McBride, Computer program for calculation of complex equilibrium compositions and applications, Tech. Rep. (1994) 1–55. NASA RP-1311.
- [116] G.A. Rossman, R.D. Braun, Thermogravimetric analysis of flexible thermal protection systems for thermal response modeling, AIAA 2016-1513, in: 54th AIAA Aerospace Sciences Meeting, San Diego, CA, 2016, <https://doi.org/10.2514/6.2016-1513>.